



```

name: The value of a helping hand
log: C:\Users\hansg\Documents\working\SHARE data\weighting\output_1.smcl
log type: smcl
opened on: 10 Jun 2026, 08:25:21
C:\Users\hansg\Documents\working\SHARE data\weighting
(0 observations deleted)
(6,505 real changes made)
(4,021 real changes made)
(42,854 real changes made)
(0 real changes made)
(14,421 real changes made)
(834 real changes made)
(738 real changes made)
(834 real changes made)

```

Skewness and kurtosis tests for normality

| Variable | Obs    | Pr(skewness) | Pr(kurtosis) | Joint test  |           |
|----------|--------|--------------|--------------|-------------|-----------|
|          |        |              |              | Adj chi2(2) | Prob>chi2 |
| age      | 67,854 | 0.0000       | 0.0000       | .           | .         |

Skewness and kurtosis tests for normality

| Variable | Obs    | Pr(skewness) | Pr(kurtosis) | Joint test  |           |
|----------|--------|--------------|--------------|-------------|-----------|
|          |        |              |              | Adj chi2(2) | Prob>chi2 |
| thinc    | 67,854 | 0.0000       | 0.0000       | .           | .         |

Skewness and kurtosis tests for normality

| Variable | Obs    | Pr(skewness) | Pr(kurtosis) | Joint test  |           |
|----------|--------|--------------|--------------|-------------|-----------|
|          |        |              |              | Adj chi2(2) | Prob>chi2 |
| yedu     | 67,854 | 0.0000       | 0.0000       | 85.25       | 0.0000    |

Summary for variables: id

Group variable: htype (Household type)

| htype          | N     |
|----------------|-------|
| 1 S            | 21750 |
| 1 CNRP         | 5644  |
| 1 C2R          | 40093 |
| 1 CNRP + 1 S   | 17    |
| Multi S        | 224   |
| 1 C2R + 1 S    | 99    |
| 1 C2R + 1 CNRP | 2     |
| Multi C2R      | 25    |
| Total          | 67854 |

| Household type | Freq.  | Percent | Cum.   |
|----------------|--------|---------|--------|
| 1              | 21,750 | 32.05   | 32.05  |
| 2              | 5,644  | 8.32    | 40.37  |
| 3              | 40,093 | 59.09   | 99.46  |
| 4              | 17     | 0.03    | 99.48  |
| 5              | 224    | 0.33    | 99.81  |
| 6              | 99     | 0.15    | 99.96  |
| 7              | 2      | 0.00    | 99.96  |
| 8              | 25     | 0.04    | 100.00 |
| Total          | 67,854 | 100.00  |        |

(21,750 real changes made)

(40,093 real changes made)

file **violinplot\_sup\_1.tif** saved as TIFF format

| household                       | Freq.  | Percent | Cum.   |
|---------------------------------|--------|---------|--------|
| Single person responding        | 21,750 | 32.05   | 32.05  |
| Couple, both responding         | 40,093 | 59.09   | 91.14  |
| Multiple, at least 1 responding | 6,011  | 8.86    | 100.00 |
| Total                           | 67,854 | 100.00  |        |

0 1 2

| Current job situation        | Freq.  | Percent | Cum.   |
|------------------------------|--------|---------|--------|
| Not applicable               | 834    | 1.23    | 1.23   |
| Retired                      | 54,482 | 80.29   | 81.52  |
| Employed or self-employed    | 4,227  | 6.23    | 87.75  |
| Unemployed                   | 480    | 0.71    | 88.46  |
| Permanently sick or disabled | 961    | 1.42    | 89.88  |
| Homemaker                    | 6,132  | 9.04    | 98.91  |
| Other                        | 738    | 1.09    | 100.00 |
| Total                        | 67,854 | 100.00  |        |

0 1 2 3 4 5 6

| Limitation<br>with<br>activities | Freq.  | Percent | Cum.   |
|----------------------------------|--------|---------|--------|
| Not limited                      | 33,477 | 49.34   | 49.34  |
| Limited                          | 34,377 | 50.66   | 100.00 |
| Total                            | 67,854 | 100.00  |        |

0 1

| Received help<br>from others<br>(how many) | Freq.  | Percent | Cum.   |
|--------------------------------------------|--------|---------|--------|
| None                                       | 52,623 | 77.55   | 77.55  |
| At least 1                                 | 15,231 | 22.45   | 100.00 |
| Total                                      | 67,854 | 100.00  |        |

0 1

| Given help to<br>others (how<br>many) | Freq.  | Percent | Cum.   |
|---------------------------------------|--------|---------|--------|
| None                                  | 40,347 | 59.46   | 59.46  |
| At least 1                            | 13,086 | 19.29   | 78.75  |
| Not applicable                        | 14,421 | 21.25   | 100.00 |
| Total                                 | 67,854 | 100.00  |        |

0 1 2

| Number of<br>activities<br>last year | Freq.  | Percent | Cum.   |
|--------------------------------------|--------|---------|--------|
| None                                 | 9,866  | 14.54   | 14.54  |
| At least 1                           | 57,154 | 84.23   | 98.77  |
| Not applicable                       | 834    | 1.23    | 100.00 |
| Total                                | 67,854 | 100.00  |        |

0 1 2

Panel variable: **id** (unbalanced)

Time variable: **Qyear, 2011 to 2015**, but with gaps

Delta: **1 unit**

note: **2.naly** omitted because of collinearity.

note: **Mundlak\_household\_Multiple** omitted because of collinearity.

note: **Mundlak\_cjs\_Other** omitted because of collinearity.

note: **Mundlak\_gali\_Limited** omitted because of collinearity.

note: **Mundlak\_rhfo\_At** omitted because of collinearity.

note: **Mundlak\_ghfo\_Not** omitted because of collinearity.

note: **Mundlak\_naly\_At** omitted because of collinearity.

note: **Mundlak\_naly\_Not** omitted because of collinearity.

note: **Mundlak\_i2** omitted because of collinearity.

note: **Mundlak\_i5** omitted because of collinearity.

note: **Mundlak\_i7** omitted because of collinearity.

note: **Mundlak\_i8** omitted because of collinearity.

Fitting comparison model:

Iteration 0: Log likelihood = **-97582.445**

Iteration 1: Log likelihood = **-78440.497**

Iteration 2: Log likelihood = **-77065.022**

Iteration 3: Log likelihood = **-77051.88**

Iteration 4: Log likelihood = **-77051.874**

Refining starting values:

Grid node 0: Log likelihood = **-75316.562**

Fitting full model:

Iteration 0: Log pseudolikelihood = **-75316.562**

Iteration 1: Log pseudolikelihood = **-73804.511**

Iteration 2: Log pseudolikelihood = **-73796.115**

Iteration 3: Log pseudolikelihood = **-73796.09**

Iteration 4: Log pseudolikelihood = **-73796.09**

Random-effects ordered logistic regression

Group variable: **id**

Number of obs = **67,854**

Number of groups = **28,019**

Random effects  $u_i \sim \text{Gaussian}$

Obs per group:

min = **1**

avg = **2.4**

max = **3**

Integration method: **mvaghermite**

Integration pts. = **12**

Log pseudolikelihood = **-73796.09**

Wald chi2(57) = **17689.30**

Prob > chi2 = **0.0000**

(Std. err. adjusted for **28,019** cl

> usters in **id**)

|                | sphus         | Odds ratio | Robust<br>std. err. | z     | P> z  | [95% co |
|----------------|---------------|------------|---------------------|-------|-------|---------|
| > n            |               |            |                     |       |       |         |
| > f. interval] |               |            |                     |       |       |         |
| > 4            | age           | .8098822   | .04468              | -3.82 | 0.000 | .726879 |
| > .9023632     |               |            |                     |       |       |         |
| > 9            | c.age#c.age   | 1.002126   | .0003812            | 5.58  | 0.000 | 1.00137 |
| > 1.002873     |               |            |                     |       |       |         |
| > 2            | yedu          | .9587765   | .0279186            | -1.45 | 0.148 | .905589 |
| > 1.015088     |               |            |                     |       |       |         |
|                | c.yedu#c.yedu | 1.001587   | .0013192            | 1.20  | 0.228 | .999005 |

|     |          |                                 |          |           |       |               |
|-----|----------|---------------------------------|----------|-----------|-------|---------------|
| > 2 | 1.004176 |                                 |          |           |       |               |
| >   |          | thinc                           | .9114051 | .0289477  | -2.92 | 0.003 .856398 |
| > 5 | .9699448 |                                 |          |           |       |               |
| >   |          | c.thinc#c.thinc                 | 1.029256 | .0133511  | 2.22  | 0.026 1.00341 |
| > 8 | 1.055759 |                                 |          |           |       |               |
| >   |          | gender                          |          |           |       |               |
| > 2 |          | Female                          | .966608  | .026838   | -1.22 | 0.221 .915412 |
| >   | 1.020667 |                                 |          |           |       |               |
|     |          | household                       |          |           |       |               |
| > 2 |          | Couple, both responding         | 1.120709 | .083273   | 1.53  | 0.125 .968825 |
| >   | 1.296405 |                                 |          |           |       |               |
| >   |          | Multiple, at least 1 responding | 1.125804 | .096472   | 1.38  | 0.167 .951748 |
| > 1 | 1.331692 |                                 |          |           |       |               |
| >   |          | cjs                             |          |           |       |               |
| > 2 |          | Not applicable                  | .9239116 | .1664988  | -0.44 | 0.661 .648985 |
| >   | 1.315304 |                                 |          |           |       |               |
| > 1 |          | Retired                         | 1.047065 | .068563   | 0.70  | 0.482 .920950 |
| >   | 1.19045  |                                 |          |           |       |               |
| > 6 |          | Unemployed                      | 1.147481 | .1813385  | 0.87  | 0.384 .841837 |
| >   | 1.564092 |                                 |          |           |       |               |
| >   |          | Permanently sick or disabled    | 1.50198  | .2199634  | 2.78  | 0.005 1.12721 |
| > 4 | 2.001344 |                                 |          |           |       |               |
| >   |          | Homemaker                       | 1.068161 | .0963509  | 0.73  | 0.465 .895067 |
| > 8 | 1.274727 |                                 |          |           |       |               |
| >   |          | Other                           | .832294  | .1061612  | -1.44 | 0.150 .648192 |
| > 2 | 1.068685 |                                 |          |           |       |               |
| >   |          | gali                            |          |           |       |               |
| > 5 |          | Limited                         | 3.253209 | .0936314  | 40.99 | 0.000 3.07477 |
| >   | 3.441998 |                                 |          |           |       |               |
| > 4 |          | lifesat                         | .8273346 | .0332384  | -4.72 | 0.000 .764687 |
| >   | .8951141 |                                 |          |           |       |               |
| >   |          | c.lifesat#c.lifesat             | 1.00045  | .0027967  | 0.16  | 0.872 .994983 |
| > 8 | 1.005947 |                                 |          |           |       |               |
| >   |          | naly                            |          |           |       |               |
| > 8 |          | At least 1                      | .8370573 | .0362048  | -4.11 | 0.000 .769021 |
| >   | .9111119 |                                 |          |           |       |               |
| >   |          | Not applicable                  | 1        | (omitted) |       |               |
|     |          | rhfo                            |          |           |       |               |
| > 4 |          | At least 1                      | 1.314193 | .0493933  | 7.27  | 0.000 1.22086 |
| >   | 1.414657 |                                 |          |           |       |               |
|     |          | ghfo                            |          |           |       |               |
| > 1 |          | At least 1                      | .9928548 | .0319662  | -0.22 | 0.824 .932138 |
| >   | 1.057526 |                                 |          |           |       |               |
| >   |          | Not applicable                  | 1.050024 | .03535    | 1.45  | 0.147 .982975 |

```

> 4
> 1.121645
                rhfo#ghto
        At least 1#At least 1 | .9042055 .0544484 -1.67 0.094 .803545
> 6
> 1.017475
        At least 1#Not applicable | .8185189 .0630185 -2.60 0.009 .703872
> 4
> .951839
                country
        Austria | 1.086577 .0737279 1.22 0.221 .951269
> 7
> 1.24113
                Germany | 1.92529 .1516567 8.32 0.000 1.64985
> 7
> 2.246705
                Spain | 2.734955 .2113283 13.02 0.000 2.35059
> 8
> 3.18216
                Italy | 1.934245 .1460148 8.74 0.000 1.66822
> 5
> 2.242684
                France | 1.798609 .1232641 8.57 0.000 1.57253
> 9
> 2.057179
                Denmark | .6764471 .0563361 -4.69 0.000 .57457
> 1
> .7963866
                Switzerland | 1.20488 .0891487 2.52 0.012 1.04223
> 1
> 1.392913
                Belgium | 1.027845 .0699733 0.40 0.687 .899455
> 3
> 1.17456
        Czech Republic | 1.672529 .1327447 6.48 0.000 1.43158
> 1
> 1.954032
                Slovenia | 1.915152 .157896 7.88 0.000 1.62939
> 1
> 2.251029
                Mundlak_age | 1.465713 .0952576 5.88 0.000 1.29041
> 3
> 1.664827
                Mundlak_thinc | .7774821 .0382588 -5.11 0.000 .705998
> 7
> .8562033
        Mundlak_lifesat | .7196188 .0471153 -5.03 0.000 .63295
> 4
> .8181497
                Mundlak_yedu | 1.051943 .0325751 1.64 0.102 .989996
> 3
> 1.117767
                Mundlak_age2 | .9968951 .0004435 -6.99 0.000 .996026
> 2
> .9977647
                Mundlak_thinc2 | 1.030064 .0214237 1.42 0.154 .988918
> 7
> 1.072921
        Mundlak_lifesat2 | 1.002213 .0045911 0.48 0.629 .993254
> 5
> 1.011252
                Mundlak_yedu2 | .9958543 .0013916 -2.97 0.003 .993130
> 6
> .9985856
        Mundlak_household_Single | .6954222 .0722254 -3.50 0.000 .56734
> 1
> .8524185
        Mundlak_household_Couple | .8288251 .066 -2.36 0.018 .709056
> 9

```

|     |           |                            |           |           |        |       |          |
|-----|-----------|----------------------------|-----------|-----------|--------|-------|----------|
| >   | .9688236  | Mundlak_household_Multiple | 1         | (omitted) |        |       |          |
| >   |           | Mundlak_cjs_Not            | 2.831881  | .8308175  | 3.55   | 0.000 | 1.5934   |
| > 9 | 5.032695  | Mundlak_cjs_Retired        | 1.384435  | .2717327  | 1.66   | 0.097 | .942326  |
| > 2 | 2.033966  | Mundlak_cjs_Employed       | .9339179  | .200265   | -0.32  | 0.750 | .613452  |
| > 6 | 1.421793  | Mundlak_cjs_Unemployed     | .9133444  | .2829943  | -0.29  | 0.770 | .497616  |
| > 6 | 1.676387  | Mundlak_cjs_Permanently    | 5.489449  | 1.500664  | 6.23   | 0.000 | 3.21243  |
| > 1 | 9.380451  | Mundlak_cjs_Homemaker      | 1.485806  | .3037347  | 1.94   | 0.053 | .995304  |
| > 4 | 2.218035  | Mundlak_cjs_Other          | 1         | (omitted) |        |       |          |
| >   |           | Mundlak_gali_Not           | .0947632  | .0045127  | -49.48 | 0.000 | .086318  |
| > 8 | .1040338  | Mundlak_gali_Limited       | 1         | (omitted) |        |       |          |
| >   |           | Mundlak_rhfo_None          | .5105053  | .0627345  | -5.47  | 0.000 | .401234  |
| > 7 | .649534   | Mundlak_rhfo_At            | 1         | (omitted) |        |       |          |
| >   |           | Mundlak_ghto_None          | .8205162  | .1085128  | -1.50  | 0.135 | .63316   |
| > 4 | 1.063306  | Mundlak_ghto_At            | .3188721  | .0471036  | -7.74  | 0.000 | .238713  |
| > 8 | .4259469  | Mundlak_ghto_Not           | 1         | (omitted) |        |       |          |
| >   |           | Mundlak_naly_None          | 1.403434  | .0935671  | 5.08   | 0.000 | 1.23152  |
| > 2 | 1.599343  | Mundlak_naly_At            | 1         | (omitted) |        |       |          |
| >   |           | Mundlak_naly_Not           | 1         | (omitted) |        |       |          |
| > 7 |           | Mundlak_i1                 | 1.211167  | .169328   | 1.37   | 0.171 | .920875  |
| >   | 1.592967  | Mundlak_i2                 | 1         | (omitted) |        |       |          |
| > 8 |           | Mundlak_i4                 | 2.071095  | .3354882  | 4.49   | 0.000 | 1.50770  |
| >   | 2.845003  | Mundlak_i5                 | 1         | (omitted) |        |       |          |
|     |           | Mundlak_i7                 | 1         | (omitted) |        |       |          |
|     |           | Mundlak_i8                 | 1         | (omitted) |        |       |          |
|     |           |                            |           |           |        |       |          |
| > 6 |           | /cut1                      | -2.639838 | 1.31487   |        |       | -5.21693 |
| >   | -.0627407 | /cut2                      | -.1546081 | 1.314917  |        |       | -2.73179 |
| > 8 | 2.422582  | /cut3                      | 3.253135  | 1.314972  |        |       | .675837  |
| > 2 | 5.830434  | /cut4                      | 6.45609   | 1.314999  |        |       | 3.8787   |
| > 4 | 9.033439  |                            |           |           |        |       |          |
|     |           |                            |           |           |        |       |          |
| > 6 |           | /sigma2_u                  | 2.308408  | .0544716  |        |       | 2.20407  |
| >   | 2.417677  |                            |           |           |        |       |          |

Note: Estimates are transformed only in the first equation to odds ratios.  
results\_impl\_xtoPlus\_sup\_1.xls

dir : seeout

(File xtoPlus\_sup\_1.doc already exists, option **append** was assumed)

```
( 1) [sphus]Mundlak_age = 0
( 2) [sphus]Mundlak_thinc = 0
( 3) [sphus]Mundlak_lifesat = 0
( 4) [sphus]Mundlak_yedu = 0
( 5) [sphus]Mundlak_age2 = 0
( 6) [sphus]Mundlak_thinc2 = 0
( 7) [sphus]Mundlak_lifesat2 = 0
( 8) [sphus]Mundlak_yedu2 = 0
( 9) [sphus]Mundlak_household_Single = 0
(10) [sphus]Mundlak_household_Couple = 0
(11) [sphus]o.Mundlak_household_Multiple = 0
(12) [sphus]Mundlak_cjs_Not = 0
(13) [sphus]Mundlak_cjs_Retired = 0
(14) [sphus]Mundlak_cjs_Employed = 0
(15) [sphus]Mundlak_cjs_Unemployed = 0
(16) [sphus]Mundlak_cjs_Permanently = 0
(17) [sphus]Mundlak_cjs_Homemaker = 0
(18) [sphus]o.Mundlak_cjs_Other = 0
(19) [sphus]Mundlak_gali_Not = 0
(20) [sphus]o.Mundlak_gali_Limited = 0
(21) [sphus]Mundlak_rhfo_None = 0
(22) [sphus]o.Mundlak_rhfo_At = 0
(23) [sphus]Mundlak_ghfo_None = 0
(24) [sphus]Mundlak_ghfo_At = 0
(25) [sphus]o.Mundlak_ghfo_Not = 0
(26) [sphus]Mundlak_naly_None = 0
(27) [sphus]o.Mundlak_naly_At = 0
(28) [sphus]o.Mundlak_naly_Not = 0
(29) [sphus]Mundlak_i1 = 0
(30) [sphus]o.Mundlak_i2 = 0
(31) [sphus]Mundlak_i4 = 0
(32) [sphus]o.Mundlak_i5 = 0
(33) [sphus]o.Mundlak_i7 = 0
(34) [sphus]o.Mundlak_i8 = 0
```

```
Constraint 11 dropped
Constraint 18 dropped
Constraint 20 dropped
Constraint 22 dropped
Constraint 25 dropped
Constraint 27 dropped
Constraint 28 dropped
Constraint 30 dropped
Constraint 32 dropped
Constraint 33 dropped
Constraint 34 dropped
```

```
chi2( 23) = 3871.59
Prob > chi2 = 0.0000
```

Click to Open File: xtoPlus\_sup\_1.doc

```
note: 2.naly omitted because of collinearity.
note: Mundlak_household_Multiple omitted because of collinearity.
note: Mundlak_cjs_Other omitted because of collinearity.
note: Mundlak_gali_Limited omitted because of collinearity.
note: Mundlak_rhfo_At omitted because of collinearity.
note: Mundlak_ghfo_Not omitted because of collinearity.
note: Mundlak_naly_At omitted because of collinearity.
note: Mundlak_naly_Not omitted because of collinearity.
note: Mundlak_i2 omitted because of collinearity.
note: Mundlak_i5 omitted because of collinearity.
note: Mundlak_i7 omitted because of collinearity.
note: Mundlak_i8 omitted because of collinearity.
```

Fitting comparison model:

```

Iteration 0: Log likelihood = -2.470e+08
Iteration 1: Log likelihood = -1.781e+08
Iteration 2: Log likelihood = -1.699e+08
Iteration 3: Log likelihood = -1.698e+08
Iteration 4: Log likelihood = -1.698e+08
Iteration 5: Log likelihood = -1.698e+08

```

Refining starting values:

```
Grid node 0: Log likelihood = -1.665e+08
```

Fitting full model:

```

Iteration 0: Log pseudolikelihood = -1.665e+08
Iteration 1: Log pseudolikelihood = -1.635e+08
Iteration 2: Log pseudolikelihood = -1.635e+08
Iteration 3: Log pseudolikelihood = -1.635e+08
Iteration 4: Log pseudolikelihood = -1.635e+08

```

Random-effects ordered logistic regression  
Group variable: **id**

Random effects  $u_i \sim \text{Gaussian}$

Number of obs = **67,842**  
Number of groups = **28,013**

Obs per group:  
min = **1**  
avg = **2.4**  
max = **3**

Integration method: **mvaghermite**

Integration pts. = **12**

Log pseudolikelihood = **-1.635e+08**

Wald chi2(57) = **7366.87**  
Prob > chi2 = **0.0000**

(Std. err. adjusted for **28,013** cl

> usters in **id**)

|                | sphus           | Odds ratio | Robust<br>std. err. | z     | P> z  | [95% co |
|----------------|-----------------|------------|---------------------|-------|-------|---------|
| > n            |                 |            |                     |       |       |         |
| > f. interval] |                 |            |                     |       |       |         |
|                | age             | .8288422   | .082354             | -1.89 | 0.059 | .682175 |
| > 7            |                 |            |                     |       |       |         |
| > 1.007042     |                 |            |                     |       |       |         |
|                | c.age#c.age     | 1.001891   | .0006927            | 2.73  | 0.006 | 1.00053 |
| > 5            |                 |            |                     |       |       |         |
| > 1.00325      |                 |            |                     |       |       |         |
|                | yedu            | .9937199   | .0414273            | -0.15 | 0.880 | .915752 |
| > 5            |                 |            |                     |       |       |         |
| > 1.078325     |                 |            |                     |       |       |         |
|                | c.yedu#c.yedu   | .9993098   | .0019867            | -0.35 | 0.728 | .995423 |
| > 6            |                 |            |                     |       |       |         |
| > 1.003211     |                 |            |                     |       |       |         |
|                | thinc           | .8703398   | .0459526            | -2.63 | 0.009 | .784777 |
| > 9            |                 |            |                     |       |       |         |
| > .9652303     |                 |            |                     |       |       |         |
|                | c.thinc#c.thinc | 1.054028   | .023729             | 2.34  | 0.019 | 1.00853 |
| > 1            |                 |            |                     |       |       |         |
| > 1.101577     |                 |            |                     |       |       |         |
|                | gender          |            |                     |       |       |         |
| > 3            | Female          | 1.01207    | .0470038            | 0.26  | 0.796 | .924013 |
| > 1.108519     |                 |            |                     |       |       |         |
|                | household       |            |                     |       |       |         |

|     |                                    |          |          |       |       |         |
|-----|------------------------------------|----------|----------|-------|-------|---------|
| > 9 | Couple, both responding            | 1.048629 | .1430811 | 0.35  | 0.728 | .802562 |
| >   | 1.37014                            |          |          |       |       |         |
| > 1 | Multiple, at least 1 responding    | 1.035045 | .1594463 | 0.22  | 0.823 | .765303 |
| >   | 1.399861                           |          |          |       |       |         |
| > 6 | cjs<br>Not applicable              | .7340685 | .2594827 | -0.87 | 0.382 | .367154 |
| >   | 1.467656                           |          |          |       |       |         |
| > 8 | Retired                            | .9915275 | .1116767 | -0.08 | 0.940 | .795120 |
| >   | 1.23645                            |          |          |       |       |         |
| > 6 | Unemployed                         | 1.22239  | .2791816 | 0.88  | 0.379 | .781274 |
| >   | 1.912562                           |          |          |       |       |         |
| > 2 | Permanently sick or disabled       | 1.852387 | .4088036 | 2.79  | 0.005 | 1.20193 |
| >   | 2.854852                           |          |          |       |       |         |
| > 7 | Homemaker                          | .958014  | .1433329 | -0.29 | 0.774 | .714528 |
| >   | 1.28447                            |          |          |       |       |         |
| > 9 | Other                              | .7531765 | .1556784 | -1.37 | 0.170 | .502292 |
| >   | 1.129371                           |          |          |       |       |         |
| > 3 | gali<br>Limited                    | 3.47111  | .1706657 | 25.31 | 0.000 | 3.15222 |
| >   | 3.822257                           |          |          |       |       |         |
| > 8 | lifesat                            | .771953  | .0539275 | -3.71 | 0.000 | .673173 |
| >   | .8852267                           |          |          |       |       |         |
| > 9 | c.lifesat#c.lifesat                | 1.002679 | .0048636 | 0.55  | 0.581 | .993191 |
| >   | 1.012257                           |          |          |       |       |         |
| > 8 | naly<br>At least 1                 | .8495872 | .0526281 | -2.63 | 0.009 | .752453 |
| >   | .9592593                           |          |          |       |       |         |
| > 7 | rhfo<br>At least 1                 | 1.192761 | .0813958 | 2.58  | 0.010 | 1.04343 |
| >   | 1.363455                           |          |          |       |       |         |
| > 5 | ghfo<br>At least 1                 | .9362337 | .0546004 | -1.13 | 0.259 | .835108 |
| >   | 1.049604                           |          |          |       |       |         |
| > 1 | Not applicable                     | 1.009548 | .053235  | 0.18  | 0.857 | .910420 |
| >   | 1.119469                           |          |          |       |       |         |
| > 8 | rhfo#ghfo<br>At least 1#At least 1 | .9220708 | .1091571 | -0.69 | 0.493 | .731133 |
| >   | 1.162872                           |          |          |       |       |         |
| > 7 | At least 1#Not applicable          | 1.068173 | .1387742 | 0.51  | 0.612 | .828048 |
| >   | 1.377931                           |          |          |       |       |         |
| > 4 | country<br>Austria                 | 1.068849 | .0785243 | 0.91  | 0.365 | .925511 |
| >   | 1.234386                           |          |          |       |       |         |
| > 8 | Germany                            | 2.012697 | .1734036 | 8.12  | 0.000 | 1.69997 |

|     |          |                            |  |          |           |       |         |
|-----|----------|----------------------------|--|----------|-----------|-------|---------|
| >   | 2.382942 |                            |  |          |           |       |         |
| >   |          | Spain                      |  | 3.242906 | .2938749  | 12.98 | 0.000   |
| > 5 |          |                            |  |          |           |       | 2.71517 |
| >   | 3.87321  |                            |  |          |           |       |         |
| >   |          | Italy                      |  | 2.156575 | .1837831  | 9.02  | 0.000   |
| > 2 |          |                            |  |          |           |       | 1.82484 |
| >   | 2.548613 |                            |  |          |           |       |         |
| >   |          | France                     |  | 1.915369 | .1439475  | 8.65  | 0.000   |
| > 3 |          |                            |  |          |           |       | 1.65303 |
| >   | 2.219339 |                            |  |          |           |       |         |
| >   |          | Denmark                    |  | .7167689 | .0629605  | -3.79 | 0.000   |
| > 7 |          |                            |  |          |           |       | .603406 |
| >   | .8514285 |                            |  |          |           |       |         |
| >   |          | Switzerland                |  | 1.149524 | .0974186  | 1.64  | 0.100   |
| > 1 |          |                            |  |          |           |       | .973602 |
| >   | 1.357234 |                            |  |          |           |       |         |
| >   |          | Belgium                    |  | 1.082277 | .0800641  | 1.07  | 0.285   |
| > 9 |          |                            |  |          |           |       | .936199 |
| >   | 1.251146 |                            |  |          |           |       |         |
| >   |          | Czech Republic             |  | 1.748698 | .1778754  | 5.49  | 0.000   |
| > 2 |          |                            |  |          |           |       | 1.43262 |
| >   | 2.134509 |                            |  |          |           |       |         |
| >   |          | Slovenia                   |  | 2.157719 | .2282058  | 7.27  | 0.000   |
| > 8 |          |                            |  |          |           |       | 1.75375 |
| >   | 2.654728 |                            |  |          |           |       |         |
| >   |          | Mundlak_age                |  | 1.545525 | .181692   | 3.70  | 0.000   |
| > 4 |          |                            |  |          |           |       | 1.22746 |
| >   | 1.946002 |                            |  |          |           |       |         |
| >   |          | Mundlak_thinc              |  | .8570858 | .069333   | -1.91 | 0.057   |
| > 8 |          |                            |  |          |           |       | .731420 |
| >   | 1.004341 |                            |  |          |           |       |         |
| >   |          | Mundlak_lifesat            |  | .7762159 | .0837142  | -2.35 | 0.019   |
| > 5 |          |                            |  |          |           |       | .628320 |
| >   | .9589233 |                            |  |          |           |       |         |
| >   |          | Mundlak_yedu               |  | .9948976 | .0443296  | -0.11 | 0.909   |
| > 9 |          |                            |  |          |           |       | .911698 |
| >   | 1.085689 |                            |  |          |           |       |         |
| >   |          | Mundlak_age2               |  | .9965936 | .0008081  | -4.21 | 0.000   |
| > 1 |          |                            |  |          |           |       | .99501  |
| >   | .9981788 |                            |  |          |           |       |         |
| >   |          | Mundlak_thinc2             |  | .9943532 | .0360344  | -0.16 | 0.876   |
| > 7 |          |                            |  |          |           |       | .92617  |
| >   | 1.067548 |                            |  |          |           |       |         |
| >   |          | Mundlak_lifesat2           |  | .999457  | .0075559  | -0.07 | 0.943   |
| > 9 |          |                            |  |          |           |       | .984756 |
| >   | 1.014376 |                            |  |          |           |       |         |
| >   |          | Mundlak_yedu2              |  | .9987392 | .0021036  | -0.60 | 0.549   |
| > 7 |          |                            |  |          |           |       | .994624 |
| >   | 1.002871 |                            |  |          |           |       |         |
| >   |          | Mundlak_household_Single   |  | .6637298 | .1190122  | -2.29 | 0.022   |
| > 6 |          |                            |  |          |           |       | .467050 |
| >   | .9432324 |                            |  |          |           |       |         |
| >   |          | Mundlak_household_Couple   |  | .7898764 | .1014071  | -1.84 | 0.066   |
| > 2 |          |                            |  |          |           |       | .614156 |
| >   | 1.015873 |                            |  |          |           |       |         |
| >   |          | Mundlak_household_Multiple |  | 1        | (omitted) |       |         |
| > 6 |          | Mundlak_cjs_Not            |  | 4.486996 | 2.644586  | 2.55  | 0.011   |
| >   | 14.24441 |                            |  |          |           |       | 1.41340 |
| >   |          | Mundlak_cjs_Retired        |  | 2.157036 | .73308    | 2.26  | 0.024   |
| > 2 |          |                            |  |          |           |       | 1.10808 |
| >   | 4.198969 |                            |  |          |           |       |         |
| >   |          | Mundlak_cjs_Employed       |  | 1.432484 | .5266361  | 0.98  | 0.328   |
| > 2 |          |                            |  |          |           |       | .696876 |
| >   | 2.944585 |                            |  |          |           |       |         |
| >   |          | Mundlak_cjs_Unemployed     |  | .817192  | .4049358  | -0.41 | 0.684   |
| > 1 |          |                            |  |          |           |       | .30941  |
| >   | 2.158303 |                            |  |          |           |       |         |
| >   |          | Mundlak_cjs_Permanently    |  | 4.455199 | 1.890844  | 3.52  | 0.000   |
| > 7 |          |                            |  |          |           |       | 1.93912 |
| >   | 10.23595 |                            |  |          |           |       |         |

|       |                       |          |           |        |       |          |
|-------|-----------------------|----------|-----------|--------|-------|----------|
| > 4   | Mundlak_cjs_Homemaker | 2.154052 | .7474235  | 2.21   | 0.027 | 1.09119  |
| >     | 4.252166              |          |           |        |       |          |
|       | Mundlak_cjs_Other     | 1        | (omitted) |        |       |          |
| > 4   | Mundlak_gali_Not      | .1028975 | .0081795  | -28.61 | 0.000 | .088052  |
| >     | .1202454              |          |           |        |       |          |
|       | Mundlak_gali_Limited  | 1        | (omitted) |        |       |          |
| > 3   | Mundlak_rhfo_None     | .5488639 | .1069928  | -3.08  | 0.002 | .374572  |
| >     | .8042547              |          |           |        |       |          |
|       | Mundlak_rhfo_At       | 1        | (omitted) |        |       |          |
| > 3   | Mundlak_ghfo_None     | 1.11039  | .2367975  | 0.49   | 0.623 | .731057  |
| >     | 1.686551              |          |           |        |       |          |
|       | Mundlak_ghfo_At       | .299488  | .0804426  | -4.49  | 0.000 | .176907  |
| > 5   |                       |          |           |        |       |          |
| >     | .5070055              |          |           |        |       |          |
|       | Mundlak_ghfo_Not      | 1        | (omitted) |        |       |          |
| > 3   | Mundlak_naly_None     | 1.215418 | .1180504  | 2.01   | 0.045 | 1.00473  |
| >     | 1.470282              |          |           |        |       |          |
|       | Mundlak_naly_At       | 1        | (omitted) |        |       |          |
|       | Mundlak_naly_Not      | 1        | (omitted) |        |       |          |
| > 5   | Mundlak_i1            | .8449105 | .1916724  | -0.74  | 0.458 | .541640  |
| >     | 1.317984              |          |           |        |       |          |
|       | Mundlak_i2            | 1        | (omitted) |        |       |          |
| > 6   | Mundlak_i4            | 2.268148 | .6519248  | 2.85   | 0.004 | 1.2912   |
| >     | 3.984087              |          |           |        |       |          |
|       | Mundlak_i5            | 1        | (omitted) |        |       |          |
|       | Mundlak_i7            | 1        | (omitted) |        |       |          |
|       | Mundlak_i8            | 1        | (omitted) |        |       |          |
| <hr/> |                       |          |           |        |       |          |
| > 3   | /cut1                 | .5052222 | 2.263157  |        |       | -3.93048 |
| >     | 4.940928              |          |           |        |       |          |
|       | /cut2                 | 2.744817 | 2.264865  |        |       | -1.69423 |
| > 6   |                       |          |           |        |       |          |
| >     | 7.183871              |          |           |        |       |          |
|       | /cut3                 | 6.25409  | 2.264233  |        |       | 1.81627  |
| > 5   |                       |          |           |        |       |          |
| >     | 10.69191              |          |           |        |       |          |
|       | /cut4                 | 9.596302 | 2.263579  |        |       | 5.15976  |
| > 9   |                       |          |           |        |       |          |
| >     | 14.03284              |          |           |        |       |          |
| <hr/> |                       |          |           |        |       |          |
| > 9   | /sigma2_u             | 2.114366 | .0861848  |        |       | 1.95201  |
| >     | 2.290216              |          |           |        |       |          |
| <hr/> |                       |          |           |        |       |          |

Note: Estimates are transformed only in the first equation to odds ratios.  
results\_imp1\_xtoPlus2\_sup\_1.xls

dir : seeout

(File xtoPlus2\_sup\_1.doc already exists, option **append** was assumed)

- ( 1) [sphus]Mundlak\_age = 0
- ( 2) [sphus]Mundlak\_thinc = 0
- ( 3) [sphus]Mundlak\_lifesat = 0
- ( 4) [sphus]Mundlak\_yedu = 0
- ( 5) [sphus]Mundlak\_age2 = 0
- ( 6) [sphus]Mundlak\_thinc2 = 0
- ( 7) [sphus]Mundlak\_lifesat2 = 0
- ( 8) [sphus]Mundlak\_yedu2 = 0
- ( 9) [sphus]Mundlak\_household\_Single = 0
- (10) [sphus]Mundlak\_household\_Couple = 0
- (11) [sphus]o.Mundlak\_household\_Multiple = 0
- (12) [sphus]Mundlak\_cjs\_Not = 0

```

(13) [sphus]Mundlak_cjs_Retired = 0
(14) [sphus]Mundlak_cjs_Employed = 0
(15) [sphus]Mundlak_cjs_Unemployed = 0
(16) [sphus]Mundlak_cjs_Permanently = 0
(17) [sphus]Mundlak_cjs_Homemaker = 0
(18) [sphus]o.Mundlak_cjs_Other = 0
(19) [sphus]Mundlak_gali_Not = 0
(20) [sphus]o.Mundlak_gali_Limited = 0
(21) [sphus]Mundlak_rhfo_None = 0
(22) [sphus]o.Mundlak_rhfo_At = 0
(23) [sphus]Mundlak_ghfo_None = 0
(24) [sphus]Mundlak_ghfo_At = 0
(25) [sphus]o.Mundlak_ghfo_Not = 0
(26) [sphus]Mundlak_naly_None = 0
(27) [sphus]o.Mundlak_naly_At = 0
(28) [sphus]o.Mundlak_naly_Not = 0
(29) [sphus]Mundlak_i1 = 0
(30) [sphus]o.Mundlak_i2 = 0
(31) [sphus]Mundlak_i4 = 0
(32) [sphus]o.Mundlak_i5 = 0
(33) [sphus]o.Mundlak_i7 = 0
(34) [sphus]o.Mundlak_i8 = 0

```

```

Constraint 11 dropped
Constraint 18 dropped
Constraint 20 dropped
Constraint 22 dropped
Constraint 25 dropped
Constraint 27 dropped
Constraint 28 dropped
Constraint 30 dropped
Constraint 32 dropped
Constraint 33 dropped
Constraint 34 dropped

```

```

chi2( 23) = 1281.12
Prob > chi2 = 0.0000

```

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note: 2.naly omitted because of collinearity.

```

Iteration 0: Log pseudolikelihood = -10538280 (not concave)
Iteration 1: Log pseudolikelihood = -4898905.8 (not concave)
Iteration 2: Log pseudolikelihood = -3647326.1 (not concave)
Iteration 3: Log pseudolikelihood = -2952947.8 (not concave)
Iteration 4: Log pseudolikelihood = -2758619.5 (not concave)
Iteration 5: Log pseudolikelihood = -2680641.8 (not concave)
Iteration 6: Log pseudolikelihood = -2636737.5 (not concave)
Iteration 7: Log pseudolikelihood = -2602361.4 (not concave)
Iteration 8: Log pseudolikelihood = -2586140.1 (not concave)
Iteration 9: Log pseudolikelihood = -2578016.3 (not concave)
Iteration 10: Log pseudolikelihood = -2549813.1 (not concave)
Iteration 11: Log pseudolikelihood = -2537846.3 (not concave)
Iteration 12: Log pseudolikelihood = -2530497.2 (not concave)
Iteration 13: Log pseudolikelihood = -2521612.8 (not concave)
Iteration 14: Log pseudolikelihood = -2518826.6 (not concave)
Iteration 15: Log pseudolikelihood = -2516322.4 (not concave)
Iteration 16: Log pseudolikelihood = -2496930.2 (not concave)
Iteration 17: Log pseudolikelihood = -2489331.9 (not concave)
Iteration 18: Log pseudolikelihood = -2487957.7 (not concave)
Iteration 19: Log pseudolikelihood = -2487437.4 (not concave)
Iteration 20: Log pseudolikelihood = -2487367.1 (not concave)
Iteration 21: Log pseudolikelihood = -2487366 (not concave)
Iteration 22: Log pseudolikelihood = -2487366 (not concave)

```

Stereotype logistic regression

Log pseudolikelihood = -2487366

Number of obs = 28,019

Wald chi2(34) = 4249.97

Prob > chi2 = 0.0000

( 1) [phi1\_1]\_cons = 1

|                |           | sphus                                | Coefficient | Robust<br>std. err. | z      | P> z  | [95% con  |
|----------------|-----------|--------------------------------------|-------------|---------------------|--------|-------|-----------|
| > f. interval] |           |                                      |             |                     |        |       |           |
| >              | .3692173  | age                                  | .2148335    | .0787687            | 2.73   | 0.006 | .0604496  |
| >              | .0000444  | c.age#c.age                          | -.0010137   | .0005399            | -1.88  | 0.060 | -.0020719 |
| >              | .0472119  | yedu                                 | -.0031735   | .0257073            | -0.12  | 0.902 | -.0535588 |
| >              | -.002862  | c.yedu#c.yedu                        | -.0051234   | .0011538            | -4.44  | 0.000 | -.0073848 |
| >              | -.4490719 | thinc                                | -.5852648   | .0694874            | -8.42  | 0.000 | -.7214576 |
| >              | .161955   | c.thinc#c.thinc                      | .1026622    | .030252             | 3.39   | 0.001 | .0433693  |
| >              | .1259261  | gender<br>Female                     | -.0040802   | .0663309            | -0.06  | 0.951 | -.1340864 |
| >              | .635746   | household<br>Couple, both responding | .4629768    | .0881491            | 5.25   | 0.000 | .2902077  |
| >              | .9856497  | Multiple, at least 1 responding      | .74791      | .121298             | 6.17   | 0.000 | .5101702  |
| >              | 1.396866  | cjs<br>Not applicable                | .6333977    | .389532             | 1.63   | 0.104 | -.130071  |
| >              | .9336907  | Retired                              | .6779463    | .1304842            | 5.20   | 0.000 | .4222019  |
| >              | 1.213278  | Unemployed                           | .6162263    | .3046238            | 2.02   | 0.043 | .0191747  |
| >              | 4.124659  | Permanently sick or disabled         | 3.573332    | .2812945            | 12.70  | 0.000 | 3.022005  |
| >              | 1.182565  | Homemaker                            | .8651914    | .1619285            | 5.34   | 0.000 | .5478174  |
| >              | .2152906  | Other                                | -.4328771   | .3307039            | -1.31  | 0.191 | -1.081045 |
| >              | 5.34761   | gali<br>Limited                      | 5.146019    | .1028542            | 50.03  | 0.000 | 4.944429  |
| >              | -.467448  | lifesat                              | -.6862247   | .1116228            | -6.15  | 0.000 | -.9050014 |
| >              | .0095786  | c.lifesat#c.lifesat                  | -.0055606   | .0077243            | -0.72  | 0.472 | -.0206999 |
| >              | -.8337727 | naly<br>At least 1                   | -1.01982    | .0949237            | -10.74 | 0.000 | -1.205867 |
|                |           | Not applicable                       | 0           | (omitted)           |        |       |           |
| >              | 1.29818   | rhfo<br>At least 1                   | 1.086585    | .1079586            | 10.06  | 0.000 | .8749902  |
| >              | -.1975534 | ghfo<br>At least 1                   | -.3866236   | .0964662            | -4.01  | 0.000 | -.5756938 |
|                |           | Not applicable                       | -.0009902   | .0834534            | -0.01  | 0.991 | -.1645558 |

|       |           |                           |           |                |       |           |
|-------|-----------|---------------------------|-----------|----------------|-------|-----------|
| >     | .1625754  |                           |           |                |       |           |
|       |           | rhfo#ghto                 |           |                |       |           |
|       |           | At least 1#At least 1     | -.6927599 | .1926877       | -3.60 | 0.000     |
| >     | -.3150991 |                           |           |                |       | -1.070421 |
|       |           | At least 1#Not applicable | -.3490592 | .1823012       | -1.91 | 0.056     |
| >     | .0082445  |                           |           |                |       | -.7063629 |
|       |           | country                   |           |                |       |           |
|       |           | Austria                   | -.2483863 | .1684737       | -1.47 | 0.140     |
| >     | .081816   |                           |           |                |       | -.5785886 |
|       |           | Germany                   | .9860791  | .1916624       | 5.14  | 0.000     |
| >     | 1.36173   |                           |           |                |       | .6104277  |
|       |           | Spain                     | 1.571972  | .1887624       | 8.33  | 0.000     |
| >     | 1.94194   |                           |           |                |       | 1.202005  |
|       |           | Italy                     | .7508422  | .1826631       | 4.11  | 0.000     |
| >     | 1.108855  |                           |           |                |       | .3928292  |
|       |           | France                    | .7911126  | .1678138       | 4.71  | 0.000     |
| >     | 1.120022  |                           |           |                |       | .4622036  |
|       |           | Denmark                   | -1.323959 | .2100707       | -6.30 | 0.000     |
| >     | -.9122281 |                           |           |                |       | -1.73569  |
|       |           | Switzerland               | -.4769901 | .1852237       | -2.58 | 0.010     |
| >     | -.1139584 |                           |           |                |       | -.8400218 |
|       |           | Belgium                   | -.1586358 | .1696728       | -0.93 | 0.350     |
| >     | .1739168  |                           |           |                |       | -.4911883 |
|       |           | Czech Republic            | .6462002  | .1844383       | 3.50  | 0.000     |
| >     | 1.007693  |                           |           |                |       | .2847077  |
|       |           | Slovenia                  | 1.308822  | .1977966       | 6.62  | 0.000     |
| >     | 1.696496  |                           |           |                |       | .921148   |
| <hr/> |           |                           |           |                |       |           |
|       |           | /phi1_1                   | 1         | (constrained)  |       |           |
|       |           | /phi1_2                   | .823838   | .0120491       | 68.37 | 0.000     |
| >     | .8474539  |                           |           |                |       | .8002222  |
|       |           | /phi1_3                   | .5772535  | .009355        | 61.71 | 0.000     |
| >     | .595589   |                           |           |                |       | .558918   |
|       |           | /phi1_4                   | .3180391  | .0083491       | 38.09 | 0.000     |
| >     | .334403   |                           |           |                |       | .3016752  |
|       |           | /phi1_5                   | 0         | (base outcome) |       |           |
| <hr/> |           |                           |           |                |       |           |
|       |           | /theta1                   | 6.524757  | 2.878616       | 2.27  | 0.023     |
| >     | 12.16674  |                           |           |                |       | .8827739  |
|       |           | /theta2                   | 6.889065  | 2.376115       | 2.90  | 0.004     |
| >     | 11.54617  |                           |           |                |       | 2.231965  |
|       |           | /theta3                   | 6.491985  | 1.670502       | 3.89  | 0.000     |
| >     | 9.766109  |                           |           |                |       | 3.217862  |
|       |           | /theta4                   | 4.233263  | .9292166       | 4.56  | 0.000     |
| >     | 6.054494  |                           |           |                |       | 2.412032  |
|       |           | /theta5                   | 0         | (base outcome) |       |           |

(sphus=Poor is the base outcome)  
 results\_impl\_slo\_sup\_1.xls  
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Predictive margins  
 Model VCE: Robust

Number of obs = 28,019

```
1. _predict: Pr(sphus==Excellent), predict(pr outcome(#1))
2. _predict: Pr(sphus==Very good), predict(pr outcome(#2))
3. _predict: Pr(sphus==Good), predict(pr outcome(#3))
4. _predict: Pr(sphus==Fair), predict(pr outcome(#4))
5. _predict: Pr(sphus==Poor), predict(pr outcome(#5))
```

|                                   | Margin   | Delta-method<br>std. err. | z      | P> z  | [95% conf. i |
|-----------------------------------|----------|---------------------------|--------|-------|--------------|
| > nterval]                        |          |                           |        |       |              |
| _predict#rhfo#ghfo<br>1#None#None | .0638093 | .0015738                  | 40.54  | 0.000 | .0607246     |
| > .066894                         |          |                           |        |       |              |
| 1#None#At least 1                 | .0718804 | .0022564                  | 31.86  | 0.000 | .067458      |
| > .0763028                        |          |                           |        |       |              |
| 1#None#Not applicable             | .0638291 | .0017887                  | 35.68  | 0.000 | .0603233     |
| > .0673349                        |          |                           |        |       |              |
| 1#At least 1#None                 | .0445629 | .0019227                  | 23.18  | 0.000 | .0407945     |
| > .0483314                        |          |                           |        |       |              |
| 1#At least 1#At least 1           | .0636652 | .0031511                  | 20.20  | 0.000 | .0574892     |
| > .0698413                        |          |                           |        |       |              |
| 1#At least 1#Not applicable       | .0502266 | .0026594                  | 18.89  | 0.000 | .0450142     |
| > .0554391                        |          |                           |        |       |              |
| 2#None#None                       | .1644944 | .0023602                  | 69.70  | 0.000 | .1598685     |
| > .1691203                        |          |                           |        |       |              |
| 2#None#At least 1                 | .1761905 | .0032819                  | 53.69  | 0.000 | .1697582     |
| > .1826228                        |          |                           |        |       |              |
| 2#None#Not applicable             | .1645242 | .0027178                  | 60.54  | 0.000 | .1591973     |
| > .1698511                        |          |                           |        |       |              |
| 2#At least 1#None                 | .1326718 | .0033282                  | 39.86  | 0.000 | .1261487     |
| > .139195                         |          |                           |        |       |              |
| 2#At least 1#At least 1           | .1642778 | .0046772                  | 35.12  | 0.000 | .1551108     |
| > .1734449                        |          |                           |        |       |              |
| 2#At least 1#Not applicable       | .1427024 | .0044495                  | 32.07  | 0.000 | .1339816     |
| > .1514233                        |          |                           |        |       |              |
| 3#None#None                       | .3775856 | .0029328                  | 128.74 | 0.000 | .3718373     |
| > .3833338                        |          |                           |        |       |              |
| 3#None#At least 1                 | .3819355 | .0030898                  | 123.61 | 0.000 | .3758796     |
| > .3879914                        |          |                           |        |       |              |
| 3#None#Not applicable             | .3775983 | .0030116                  | 125.38 | 0.000 | .3716956     |
| > .3835009                        |          |                           |        |       |              |
| 3#At least 1#None                 | .3590447 | .0032919                  | 109.07 | 0.000 | .3525926     |
| > .3654968                        |          |                           |        |       |              |
| 3#At least 1#At least 1           | .377493  | .003362                   | 112.28 | 0.000 | .3709035     |
| > .3840825                        |          |                           |        |       |              |
| 3#At least 1#Not applicable       | .3659969 | .0037554                  | 97.46  | 0.000 | .3586365     |
| > .3733573                        |          |                           |        |       |              |
| 4#None#None                       | .2836358 | .0027822                  | 101.95 | 0.000 | .2781827     |
| > .2890889                        |          |                           |        |       |              |
| 4#None#At least 1                 | .272     | .0035092                  | 77.51  | 0.000 | .2651221     |
| > .2788779                        |          |                           |        |       |              |
| 4#None#Not applicable             | .2836068 | .003069                   | 92.41  | 0.000 | .2775917     |
| > .2896218                        |          |                           |        |       |              |
| 4#At least 1#None                 | .3126752 | .0038389                  | 81.45  | 0.000 | .3051511     |
| > .3201992                        |          |                           |        |       |              |
| 4#At least 1#At least 1           | .2838468 | .004814                   | 58.96  | 0.000 | .2744117     |
| > .293282                         |          |                           |        |       |              |
| 4#At least 1#Not applicable       | .303982  | .0045312                  | 67.09  | 0.000 | .295101      |
| > .312863                         |          |                           |        |       |              |
| 5#None#None                       | .1104749 | .0021334                  | 51.78  | 0.000 | .1062935     |
| > .1146563                        |          |                           |        |       |              |
| 5#None#At least 1                 | .0979936 | .0030349                  | 32.29  | 0.000 | .0920454     |
| > .1039418                        |          |                           |        |       |              |
| 5#None#Not applicable             | .1104416 | .0026628                  | 41.48  | 0.000 | .1052227     |
| > .1156606                        |          |                           |        |       |              |
| 5#At least 1#None                 | .1510454 | .004108                   | 36.77  | 0.000 | .1429939     |
| > .1590968                        |          |                           |        |       |              |
| 5#At least 1#At least 1           | .1107171 | .0049721                  | 22.27  | 0.000 | .1009719     |
| > .1204623                        |          |                           |        |       |              |
| 5#At least 1#Not applicable       | .137092  | .0056427                  | 24.30  | 0.000 | .1260325     |
| > .1481515                        |          |                           |        |       |              |

slogM\_sup\_1.xls  
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note: **2.naly** omitted because of collinearity.

```
Iteration 0: Log pseudolikelihood = -7978269.4 (not concave)
Iteration 1: Log pseudolikelihood = -5858707.1 (not concave)
Iteration 2: Log pseudolikelihood = -3236246.4 (not concave)
Iteration 3: Log pseudolikelihood = -2339530.6 (not concave)
Iteration 4: Log pseudolikelihood = -1947983.3 (not concave)
Iteration 5: Log pseudolikelihood = -1817399.7 (not concave)
Iteration 6: Log pseudolikelihood = -1767033 (not concave)
Iteration 7: Log pseudolikelihood = -1731500 (not concave)
Iteration 8: Log pseudolikelihood = -1712807 (not concave)
Iteration 9: Log pseudolikelihood = -1696034.3 (not concave)
Iteration 10: Log pseudolikelihood = -1688563.9 (not concave)
Iteration 11: Log pseudolikelihood = -1677226.6 (not concave)
Iteration 12: Log pseudolikelihood = -1670576.7 (not concave)
Iteration 13: Log pseudolikelihood = -1668528.2 (not concave)
Iteration 14: Log pseudolikelihood = -1666524.7 (not concave)
Iteration 15: Log pseudolikelihood = -1660877.3 (not concave)
Iteration 16: Log pseudolikelihood = -1655216.7 (not concave)
Iteration 17: Log pseudolikelihood = -1653815 (not concave)
Iteration 18: Log pseudolikelihood = -1652255.7 (not concave)
Iteration 19: Log pseudolikelihood = -1649786.8 (not concave)
Iteration 20: Log pseudolikelihood = -1647450.2 (not concave)
Iteration 21: Log pseudolikelihood = -1645733.1 (not concave)
Iteration 22: Log pseudolikelihood = -1644336.3 (not concave)
Iteration 23: Log pseudolikelihood = -1643043.4 (not concave)
Iteration 24: Log pseudolikelihood = -1640581.2 (not concave)
Iteration 25: Log pseudolikelihood = -1638873.9 (not concave)
Iteration 26: Log pseudolikelihood = -1637718.2 (not concave)
Iteration 27: Log pseudolikelihood = -1636526.1 (not concave)
Iteration 28: Log pseudolikelihood = -1625146.7 (not concave)
Iteration 29: Log pseudolikelihood = -1622400.1 (not concave)
Iteration 30: Log pseudolikelihood = -1621661.6 (not concave)
Iteration 31: Log pseudolikelihood = -1620712.4 (not concave)
Iteration 32: Log pseudolikelihood = -1620211.6 (not concave)
Iteration 33: Log pseudolikelihood = -1613740.8 (not concave)
Iteration 34: Log pseudolikelihood = -1612893.1 (not concave)
Iteration 35: Log pseudolikelihood = -1612869.9 (not concave)
Iteration 36: Log pseudolikelihood = -1612478.4 (not concave)
Iteration 37: Log pseudolikelihood = -1612342 (not concave)
Iteration 38: Log pseudolikelihood = -1612336.2 (not concave)
Iteration 39: Log pseudolikelihood = -1612336.2 (not concave)
```

Stereotype logistic regression

Number of obs = **18,549**

Wald chi2(32) = **2538.61**

Prob > chi2 = **0.0000**

Log pseudolikelihood = **-1612336.2**

( 1) **[phi1\_1]\_cons = 1**

|                |           | sphus           | Coefficient | Robust<br>std. err. | z     | P> z  | [95% con  |
|----------------|-----------|-----------------|-------------|---------------------|-------|-------|-----------|
| > f. interval] |           |                 |             |                     |       |       |           |
| >              | .4077132  | age             | .1996693    | .1061468            | 1.88  | 0.060 | -.0083745 |
| >              | .0004515  | c.age#c.age     | -.0009153   | .0006974            | -1.31 | 0.189 | -.0022822 |
| >              | .0669898  | yedu            | .0061034    | .031065             | 0.20  | 0.844 | -.054783  |
| >              | -.0022045 | c.yedu#c.yedu   | -.0049667   | .0014093            | -3.52 | 0.000 | -.0077288 |
| >              | -.4629742 | thinc           | -.6519624   | .0964243            | -6.76 | 0.000 | -.8409507 |
| >              | .2041205  | c.thinc#c.thinc | .1232459    | .0412633            | 2.99  | 0.003 | .0423713  |

|   |                                 |           |           |       |       |           |
|---|---------------------------------|-----------|-----------|-------|-------|-----------|
|   | gender                          |           |           |       |       |           |
| > | Female                          | -.0393559 | .0802571  | -0.49 | 0.624 | -.196657  |
|   | household                       |           |           |       |       |           |
| > | Couple, both responding         | .4805778  | .0959641  | 5.01  | 0.000 | .2924916  |
| > | .6686639                        |           |           |       |       |           |
| > | Multiple, at least 1 responding | .4585629  | .1617058  | 2.84  | 0.005 | .1416253  |
| > | .7755005                        |           |           |       |       |           |
|   | cjs                             |           |           |       |       |           |
| > | Not applicable                  | 2.45772   | .4131685  | 5.95  | 0.000 | 1.647924  |
| > | Retired                         | 1.313532  | .2460489  | 5.34  | 0.000 | .8312847  |
| > | Unemployed                      | .5624786  | .6927467  | 0.81  | 0.417 | -.7952799 |
| > | 1.920237                        |           |           |       |       |           |
| > | Permanently sick or disabled    | 3.712661  | .5181457  | 7.17  | 0.000 | 2.697114  |
| > | 4.728208                        |           |           |       |       |           |
| > | Homemaker                       | 1.589921  | .2806062  | 5.67  | 0.000 | 1.039942  |
| > | 2.139899                        |           |           |       |       |           |
| > | Other                           | 1.500491  | .3840386  | 3.91  | 0.000 | .7477892  |
| > | 2.253193                        |           |           |       |       |           |
|   | gali                            |           |           |       |       |           |
| > | Limited                         | 4.994458  | .1253334  | 39.85 | 0.000 | 4.748809  |
| > | lifesat                         | -.8020182 | .1373484  | -5.84 | 0.000 | -1.071216 |
| > | -.5328202                       |           |           |       |       |           |
| > | c.lifesat#c.lifesat             | .0050286  | .009449   | 0.53  | 0.595 | -.0134911 |
| > | .0235482                        |           |           |       |       |           |
|   | naly                            |           |           |       |       |           |
| > | At least 1                      | -.5524794 | .118534   | -4.66 | 0.000 | -.7848019 |
| > | -.320157                        |           |           |       |       |           |
|   | Not applicable                  | 0         | (omitted) |       |       |           |
|   | rhfo                            |           |           |       |       |           |
| > | At least 1                      | 1.181987  | .1035452  | 11.42 | 0.000 | .9790424  |
| > | 1.384932                        |           |           |       |       |           |
|   | ghfo                            |           |           |       |       |           |
| > | At least 1                      | -.666128  | .1058232  | -6.29 | 0.000 | -.8735375 |
| > | -.4587184                       |           |           |       |       |           |
|   | rhfo#ghfo                       |           |           |       |       |           |
| > | At least 1#At least 1           | -.8157391 | .1872022  | -4.36 | 0.000 | -1.182649 |
| > | -.4488294                       |           |           |       |       |           |
|   | country                         |           |           |       |       |           |
| > | Austria                         | .5038424  | .1883781  | 2.67  | 0.007 | .134628   |
| > | .8730567                        |           |           |       |       |           |
| > | Germany                         | 1.828061  | .2197499  | 8.32  | 0.000 | 1.397359  |
| > | 2.258763                        |           |           |       |       |           |
| > | Spain                           | 1.76209   | .2133341  | 8.26  | 0.000 | 1.343963  |
| > | 2.180217                        |           |           |       |       |           |
| > | Italy                           | 1.454461  | .205838   | 7.07  | 0.000 | 1.051026  |
| > | 1.857896                        |           |           |       |       |           |
| > | France                          | 1.477333  | .1898295  | 7.78  | 0.000 | 1.105274  |
| > | 1.849392                        |           |           |       |       |           |
| > | Denmark                         | -.5802786 | .2399156  | -2.42 | 0.016 | -1.050504 |
| > | -.1100527                       |           |           |       |       |           |
| > | Switzerland                     | .3582276  | .2079487  | 1.72  | 0.085 | -.0493444 |
| > | .7657996                        |           |           |       |       |           |
| > | Belgium                         | .256203   | .1845074  | 1.39  | 0.165 | -.1054248 |
| > | .6178308                        |           |           |       |       |           |
| > | Czech Republic                  | 1.852082  | .2129305  | 8.70  | 0.000 | 1.434746  |
| > | 2.269418                        |           |           |       |       |           |
| > | Slovenia                        | 1.686099  | .2249387  | 7.50  | 0.000 | 1.245227  |
| > | 2.126971                        |           |           |       |       |           |

|   |          |          | 1 (constrained) |                |       |           |
|---|----------|----------|-----------------|----------------|-------|-----------|
|   | /phi1_1  | .849094  | .0153581        | 55.29          | 0.000 | .8189926  |
| > | .8791954 | /phi1_2  |                 |                |       |           |
| > | .6244969 | /phi1_3  | .6012456        | .0118631       | 50.68 | 0.000     |
| > | .3380434 | /phi1_4  | .3178179        | .0103193       | 30.80 | 0.000     |
|   |          | /phi1_5  | 0               | (base outcome) |       |           |
|   | /theta1  | 7.507777 | 4.036116        | 1.86           | 0.063 | -.4028656 |
| > | 15.41842 | /theta2  | 7.903664        | 3.428378       | 2.31  | 0.021     |
| > | 14.62316 | /theta3  | 7.322293        | 2.434279       | 3.01  | 0.003     |
| > | 12.09339 | /theta4  | 4.615652        | 1.297485       | 3.56  | 0.000     |
| > | 7.158675 | /theta5  | 0               | (base outcome) |       |           |

(sphus=Poor is the base outcome)  
results\_impl\_slo\_sup\_1.xls  
 dir : seeout

Predictive margins  
 Model VCE: **Robust**

Number of obs = **18,549**

1.\_predict: **Pr(sphus==Excellent)**, predict(pr outcome(#1))  
 2.\_predict: **Pr(sphus==Very good)**, predict(pr outcome(#2))  
 3.\_predict: **Pr(sphus==Good)**, predict(pr outcome(#3))  
 4.\_predict: **Pr(sphus==Fair)**, predict(pr outcome(#4))  
 5.\_predict: **Pr(sphus==Poor)**, predict(pr outcome(#5))

|        |                         | Delta-method |          |        |                  |          |
|--------|-------------------------|--------------|----------|--------|------------------|----------|
|        | Margin                  | std. err.    | z        | P> z   | [95% conf. inter |          |
| > val] |                         |              |          |        |                  |          |
|        | _predict#rhfo#ghto      |              |          |        |                  |          |
|        | 1#None#None             | .051734      | .0016705 | 30.97  | 0.000            | .0484599 |
| > 5008 | 1#None#At least 1       | .063474      | .0023191 | 27.37  | 0.000            | .0589287 |
| > 0193 | 1#At least 1#None       | .0348555     | .0016666 | 20.91  | 0.000            | .0315889 |
| > 1221 | 1#At least 1#At least 1 | .0568116     | .0027585 | 20.60  | 0.000            | .0514051 |
| > 2182 | 2#None#None             | .1535602     | .0026895 | 57.10  | 0.000            | .148289  |
| > 8315 | 2#None#At least 1       | .1748507     | .0038909 | 44.94  | 0.000            | .1672247 |
| > 4768 | 2#At least 1#None       | .1183569     | .0032746 | 36.14  | 0.000            | .1119387 |
| > 4775 | 2#At least 1#At least 1 | .1630413     | .0048623 | 33.53  | 0.000            | .1535114 |
| > 5711 | 3#None#None             | .3889156     | .0036652 | 106.11 | 0.000            | .3817318 |
| > 0993 | 3#None#At least 1       | .3995749     | .0040736 | 98.09  | 0.000            | .3915909 |
| > 7559 | 3#At least 1#None       | .3611046     | .0039165 | 92.20  | 0.000            | .3534285 |
| > 7807 | 3#At least 1#At least 1 | .3941917     | .0042251 | 93.30  | 0.000            | .3859106 |
| > 4729 | 4#None#None             | .3007309     | .0033758 | 89.08  | 0.000            | .2941145 |
| > 3473 | 4#None#At least 1       | .2778493     | .0043152 | 64.39  | 0.000            | .2693916 |
| > 6307 |                         |              |          |        |                  |          |

|                         |  |          |          |       |       |          |      |
|-------------------------|--|----------|----------|-------|-------|----------|------|
| 4#At least 1#None       |  | .3357759 | .0044016 | 76.29 | 0.000 | .3271489 | .344 |
| > 4028                  |  |          |          |       |       |          |      |
| 4#At least 1#At least 1 |  | .2906575 | .0053305 | 54.53 | 0.000 | .2802099 | .301 |
| > 1051                  |  |          |          |       |       |          |      |
| 5#None#None             |  | .1050594 | .0024182 | 43.45 | 0.000 | .1003199 | .109 |
| > 7989                  |  |          |          |       |       |          |      |
| 5#None#At least 1       |  | .0842511 | .0033091 | 25.46 | 0.000 | .0777653 | .090 |
| > 7368                  |  |          |          |       |       |          |      |
| 5#At least 1#None       |  | .1499071 | .0039665 | 37.79 | 0.000 | .142133  | .157 |
| > 6813                  |  |          |          |       |       |          |      |
| 5#At least 1#At least 1 |  | .0952979 | .0045322 | 21.03 | 0.000 | .0864149 | .104 |
| > 1809                  |  |          |          |       |       |          |      |

slogM\_sup\_1.xls

dir : seeout

note: 2.naly omitted because of collinearity.

note: 11.country omitted because of collinearity.

```

Iteration 0: Log pseudolikelihood = -1275098.7 (not concave)
Iteration 1: Log pseudolikelihood = -928884.91 (not concave)
Iteration 2: Log pseudolikelihood = -777631.24 (not concave)
Iteration 3: Log pseudolikelihood = -421498.27 (not concave)
Iteration 4: Log pseudolikelihood = -264881.99 (not concave)
Iteration 5: Log pseudolikelihood = -224233.39 (not concave)
Iteration 6: Log pseudolikelihood = -202211.61 (not concave)
Iteration 7: Log pseudolikelihood = -191766.55 (not concave)
Iteration 8: Log pseudolikelihood = -190361.23 (not concave)
Iteration 9: Log pseudolikelihood = -189572.53 (not concave)
Iteration 10: Log pseudolikelihood = -189263.82 (not concave)
Iteration 11: Log pseudolikelihood = -188982.64 (not concave)
Iteration 12: Log pseudolikelihood = -188744.42 (not concave)
Iteration 13: Log pseudolikelihood = -188505.6 (not concave)
Iteration 14: Log pseudolikelihood = -188286.57 (not concave)
Iteration 15: Log pseudolikelihood = -188079.69 (not concave)
Iteration 16: Log pseudolikelihood = -187878.92 (not concave)
Iteration 17: Log pseudolikelihood = -187686.8 (not concave)
Iteration 18: Log pseudolikelihood = -187502.71 (not concave)
Iteration 19: Log pseudolikelihood = -187323.51 (not concave)
Iteration 20: Log pseudolikelihood = -187150.26 (not concave)
Iteration 21: Log pseudolikelihood = -186983.11 (not concave)
Iteration 22: Log pseudolikelihood = -186822.76 (not concave)
Iteration 23: Log pseudolikelihood = -186668.13 (not concave)
Iteration 24: Log pseudolikelihood = -186519.78 (not concave)
Iteration 25: Log pseudolikelihood = -186376.76 (not concave)
Iteration 26: Log pseudolikelihood = -186239.03 (not concave)
Iteration 27: Log pseudolikelihood = -186105.16 (not concave)
Iteration 28: Log pseudolikelihood = -185974.55 (not concave)
Iteration 29: Log pseudolikelihood = -185845.18 (not concave)
Iteration 30: Log pseudolikelihood = -185715.65 (not concave)
Iteration 31: Log pseudolikelihood = -185583.72 (not concave)
Iteration 32: Log pseudolikelihood = -185448.8 (not concave)
Iteration 33: Log pseudolikelihood = -185313.31 (not concave)
Iteration 34: Log pseudolikelihood = -185186.78 (not concave)
Iteration 35: Log pseudolikelihood = -185084.53 (not concave)
Iteration 36: Log pseudolikelihood = -184520.23 (not concave)
Iteration 37: Log pseudolikelihood = -184388.96 (not concave)
Iteration 38: Log pseudolikelihood = -184324.09 (not concave)
Iteration 39: Log pseudolikelihood = -184245.53 (not concave)
Iteration 40: Log pseudolikelihood = -184222.51 (not concave)
Iteration 41: Log pseudolikelihood = -184220.41 (not concave)
Iteration 42: Log pseudolikelihood = -184220.4 (not concave)

```

Stereotype logistic regression

Log pseudolikelihood = -184220.4

Number of obs = 2,086

Wald chi2(21) = 420.63

Prob > chi2 = 0.0000

( 1) [phi1\_1]\_cons = 1

|                |           | sphus                                | Coefficient | Robust<br>std. err. | z     | P> z  | [95% con  |
|----------------|-----------|--------------------------------------|-------------|---------------------|-------|-------|-----------|
| > f. interval] |           |                                      |             |                     |       |       |           |
| >              | .3102786  | age                                  | -.2707435   | .2964453            | -0.91 | 0.361 | -.8517656 |
| >              | .0060331  | c.age#c.age                          | .0022151    | .001948             | 1.14  | 0.255 | -.0016029 |
| >              | .1692382  | yedu                                 | .018842     | .0767342            | 0.25  | 0.806 | -.1315542 |
| >              | .0008651  | c.yedu#c.yedu                        | -.00638     | .0036966            | -1.73 | 0.084 | -.0136252 |
| >              | -.8783719 | thinc                                | -1.435056   | .2840278            | -5.05 | 0.000 | -1.99174  |
| >              | .7714338  | c.thinc#c.thinc                      | .4512668    | .1633535            | 2.76  | 0.006 | .1310998  |
| >              | .3337794  | gender<br>Female                     | -.1390083   | .2412227            | -0.58 | 0.564 | -.6117961 |
| >              | 1.4105    | household<br>Couple, both responding | .8450144    | .2885183            | 2.93  | 0.003 | .279529   |
| >              | 1.456865  | Multiple, at least 1 responding      | .3698619    | .5546036            | 0.67  | 0.505 | -.7171412 |
| >              | 7.196328  | cjs<br>Not applicable                | 3.609929    | 1.829829            | 1.97  | 0.049 | .0235301  |
| >              | 6.11476   | Retired                              | 3.231249    | 1.471206            | 2.20  | 0.028 | .3477384  |
| >              | -2.187579 | Unemployed                           | -5.886411   | 1.887194            | -3.12 | 0.002 | -9.585243 |
| >              | 13.94214  | Permanently sick or disabled         | 8.53876     | 2.75688             | 3.10  | 0.002 | 3.135375  |
| >              | 6.323696  | Homemaker                            | 3.344281    | 1.520137            | 2.20  | 0.028 | .364867   |
| >              | 5.412994  | Other                                | 1.949273    | 1.767237            | 1.10  | 0.270 | -1.514449 |
| >              | 6.311496  | gali<br>Limited                      | 5.488832    | .4197341            | 13.08 | 0.000 | 4.666169  |
| >              | .759686   | lifesat                              | -.0651969   | .4208664            | -0.15 | 0.877 | -.8900798 |
| >              | .0103162  | c.lifesat#c.lifesat                  | -.0453301   | .0283915            | -1.60 | 0.110 | -.1009764 |
| >              | -.3402694 | naly<br>At least 1                   | -1.417786   | .5497634            | -2.58 | 0.010 | -2.495302 |
|                |           | Not applicable                       | 0           | (omitted)           |       |       |           |
| >              | 1.676278  | rhfo<br>At least 1                   | 1.125999    | .2807597            | 4.01  | 0.000 | .5757198  |
| >              | -.0692504 | ghto<br>At least 1                   | -.7000928   | .3218643            | -2.18 | 0.030 | -1.330935 |

|                       |          |           |                |       |       |           |
|-----------------------|----------|-----------|----------------|-------|-------|-----------|
| rhfo#ghto             |          |           |                |       |       |           |
| At least 1#At least 1 |          | -1.869345 | .5250152       | -3.56 | 0.000 | -2.898356 |
| >                     | -.840334 |           |                |       |       |           |
| country               |          |           |                |       |       |           |
| Austria               |          | 0         | (omitted)      |       |       |           |
| <hr/>                 |          |           |                |       |       |           |
| /phi1_1               |          | 1         | (constrained)  |       |       |           |
| /phi1_2               |          | .8698828  | .0439659       | 19.79 | 0.000 | .7837112  |
| >                     | .9560543 |           |                |       |       |           |
| /phi1_3               |          | .6082704  | .0360651       | 16.87 | 0.000 | .537584   |
| >                     | .6789568 |           |                |       |       |           |
| /phi1_4               |          | .2917349  | .0364014       | 8.01  | 0.000 | .2203896  |
| >                     | .3630803 |           |                |       |       |           |
| /phi1_5               |          | 0         | (base outcome) |       |       |           |
| <hr/>                 |          |           |                |       |       |           |
| /theta1               |          | -6.476512 | 11.28718       | -0.57 | 0.566 | -28.59898 |
| >                     | 15.64595 |           |                |       |       |           |
| /theta2               |          | -4.032561 | 9.808242       | -0.41 | 0.681 | -23.25636 |
| >                     | 15.19124 |           |                |       |       |           |
| /theta3               |          | -1.191397 | 6.851065       | -0.17 | 0.862 | -14.61924 |
| >                     | 12.23644 |           |                |       |       |           |
| /theta4               |          | .3662706  | 3.286369       | 0.11  | 0.911 | -6.074894 |
| >                     | 6.807435 |           |                |       |       |           |
| /theta5               |          | 0         | (base outcome) |       |       |           |
| <hr/>                 |          |           |                |       |       |           |

(sphus=Poor is the base outcome)

results\_impl\_sloCountry\_sup\_1.xls

dir : seeout

note: 2.naly omitted because of collinearity.

note: 12.country omitted because of collinearity.

```

Iteration 0: Log pseudolikelihood = -1157201.8 (not concave)
Iteration 1: Log pseudolikelihood = -972081.18 (not concave)
Iteration 2: Log pseudolikelihood = -581169.59 (not concave)
Iteration 3: Log pseudolikelihood = -496881.01 (not concave)
Iteration 4: Log pseudolikelihood = -334428.98 (not concave)
Iteration 5: Log pseudolikelihood = -247066.7 (not concave)
Iteration 6: Log pseudolikelihood = -138535.95 (not concave)
Iteration 7: Log pseudolikelihood = -105294.03 (not concave)
Iteration 8: Log pseudolikelihood = -93415.748 (not concave)
Iteration 9: Log pseudolikelihood = -85907.527 (not concave)
Iteration 10: Log pseudolikelihood = -84008.46 (not concave)
Iteration 11: Log pseudolikelihood = -80891.752 (not concave)
Iteration 12: Log pseudolikelihood = -79336.637 (not concave)
Iteration 13: Log pseudolikelihood = -76169.352 (not concave)
Iteration 14: Log pseudolikelihood = -74652.584 (not concave)
Iteration 15: Log pseudolikelihood = -72852.805 (not concave)
Iteration 16: Log pseudolikelihood = -70763.841 (not concave)
Iteration 17: Log pseudolikelihood = -69849.788 (not concave)
Iteration 18: Log pseudolikelihood = -69034.01 (not concave)
Iteration 19: Log pseudolikelihood = -68331.922 (not concave)
Iteration 20: Log pseudolikelihood = -67647.915 (not concave)
Iteration 21: Log pseudolikelihood = -66963.228 (not concave)
Iteration 22: Log pseudolikelihood = -66325.105 (not concave)
Iteration 23: Log pseudolikelihood = -65730.742 (not concave)
Iteration 24: Log pseudolikelihood = -65174.199 (not concave)
Iteration 25: Log pseudolikelihood = -64646.124 (not concave)
Iteration 26: Log pseudolikelihood = -64166.158 (not concave)
Iteration 27: Log pseudolikelihood = -63726.787 (not concave)
Iteration 28: Log pseudolikelihood = -63328.209 (not concave)
Iteration 29: Log pseudolikelihood = -62963.817 (not concave)
Iteration 30: Log pseudolikelihood = -62639.47 (not concave)
Iteration 31: Log pseudolikelihood = -62348.828 (not concave)
Iteration 32: Log pseudolikelihood = -62092.554 (not concave)
Iteration 33: Log pseudolikelihood = -61865.874 (not concave)
Iteration 34: Log pseudolikelihood = -61669.045 (not concave)
Iteration 35: Log pseudolikelihood = -61497.413 (not concave)
Iteration 36: Log pseudolikelihood = -61349.88 (not concave)

```

```

Iteration 37: Log pseudolikelihood = -61222.322 (not concave)
Iteration 38: Log pseudolikelihood = -61113.174 (not concave)
Iteration 39: Log pseudolikelihood = -61019.017 (not concave)
Iteration 40: Log pseudolikelihood = -60938.208 (not concave)
Iteration 41: Log pseudolikelihood = -60868.243 (not concave)
Iteration 42: Log pseudolikelihood = -60807.619 (not concave)
Iteration 43: Log pseudolikelihood = -60754.683 (not concave)
Iteration 44: Log pseudolikelihood = -60708.177 (not concave)
Iteration 45: Log pseudolikelihood = -60667.063 (not concave)
Iteration 46: Log pseudolikelihood = -60630.382 (not concave)
Iteration 47: Log pseudolikelihood = -60597.473 (not concave)
Iteration 48: Log pseudolikelihood = -60567.66 (not concave)
Iteration 49: Log pseudolikelihood = -60540.504 (not concave)
Iteration 50: Log pseudolikelihood = -60515.554 (not concave)
Iteration 51: Log pseudolikelihood = -60492.51 (not concave)
Iteration 52: Log pseudolikelihood = -60471.074 (not concave)
Iteration 53: Log pseudolikelihood = -60451.04 (not concave)
Iteration 54: Log pseudolikelihood = -60432.213
Iteration 55: Log pseudolikelihood = -60364.529 (not concave)
Iteration 56: Log pseudolikelihood = -60164.069
Iteration 57: Log pseudolikelihood = -60076.829 (not concave)
Iteration 58: Log pseudolikelihood = -60032.007
Iteration 59: Log pseudolikelihood = -59996.276 (not concave)
Iteration 60: Log pseudolikelihood = -59971.154
Iteration 61: Log pseudolikelihood = -59941.842 (not concave)
Iteration 62: Log pseudolikelihood = -59884.677
Iteration 63: Log pseudolikelihood = -59863.158
Iteration 64: Log pseudolikelihood = -59845.908
Iteration 65: Log pseudolikelihood = -59842.282
Iteration 66: Log pseudolikelihood = -59839.756 (not concave)
Iteration 67: Log pseudolikelihood = -59837.222
Iteration 68: Log pseudolikelihood = -59835.119
Iteration 69: Log pseudolikelihood = -59835.025
Iteration 70: Log pseudolikelihood = -59835.024

```

Stereotype logistic regression

Number of obs = 771

Wald chi2(22) = 93.23

Prob > chi2 = 0.0000

Log pseudolikelihood = -59835.024

( 1) [phi1\_1]\_cons = 1

|                | sphus                                | Coefficient | Robust<br>std. err. | z     | P> z  | [95% con  |
|----------------|--------------------------------------|-------------|---------------------|-------|-------|-----------|
| > f. interval] |                                      |             |                     |       |       |           |
| > 1.628234     | age                                  | .5945163    | .5274166            | 1.13  | 0.260 | -.4392013 |
| > .0035201     | c.age#c.age                          | -.003259    | .0034588            | -0.94 | 0.346 | -.0100381 |
| > .185161      | yedu                                 | -.303587    | .2493658            | -1.22 | 0.223 | -.7923351 |
| > .0221303     | c.yedu#c.yedu                        | .0059319    | .0082647            | 0.72  | 0.473 | -.0102666 |
| > -.33208      | thinc                                | -1.195067   | .4403075            | -2.71 | 0.007 | -2.058054 |
| > .7149789     | c.thinc#c.thinc                      | .2896032    | .2170324            | 1.33  | 0.182 | -.1357725 |
| > .3462812     | gender<br>Female                     | -.4419995   | .4021914            | -1.10 | 0.272 | -1.23028  |
| > 2.051577     | household<br>Couple, both responding | 1.066551    | .5025738            | 2.12  | 0.034 | .0815245  |

|                                 |                              |           |                |       |       |           |
|---------------------------------|------------------------------|-----------|----------------|-------|-------|-----------|
| Multiple, at least 1 responding |                              | .6526076  | .7078067       | 0.92  | 0.357 | -.734668  |
| > 2.039883                      |                              |           |                |       |       |           |
|                                 | cjs                          |           |                |       |       |           |
| > 7.984598                      | Not applicable               | 4.499114  | 1.778341       | 2.53  | 0.011 | 1.013629  |
| > 4.377363                      | Retired                      | 2.407055  | 1.005278       | 2.39  | 0.017 | .436746   |
| > 3.007281                      | Unemployed                   | -.6105739 | 1.845878       | -0.33 | 0.741 | -4.228429 |
| > 8.403621                      | Permanently sick or disabled | 5.856612  | 1.299518       | 4.51  | 0.000 | 3.309603  |
| > 5.616275                      | Homemaker                    | 2.657456  | 1.509629       | 1.76  | 0.078 | -.3013628 |
| > 2.101716                      | Other                        | -1.102268 | 1.634716       | -0.67 | 0.500 | -4.306252 |
|                                 | gali                         |           |                |       |       |           |
| > 7.063178                      | Limited                      | 5.501752  | .7966607       | 6.91  | 0.000 | 3.940326  |
| > -.567858                      | lifesat                      | -2.296294 | .8818711       | -2.60 | 0.009 | -4.024729 |
| > .1931609                      | c.lifesat#c.lifesat          | .0814474  | .0569978       | 1.43  | 0.153 | -.0302662 |
|                                 | naly                         |           |                |       |       |           |
| > 2.518336                      | At least 1                   | 1.114806  | .7160997       | 1.56  | 0.120 | -.2887237 |
|                                 | Not applicable               | 0         | (omitted)      |       |       |           |
|                                 | rhfo                         |           |                |       |       |           |
| > 2.229316                      | At least 1                   | 1.205004  | .522618        | 2.31  | 0.021 | .1806917  |
|                                 | ghfo                         |           |                |       |       |           |
| > -.0396443                     | At least 1                   | -1.098663 | .5403255       | -2.03 | 0.042 | -2.157681 |
|                                 | rhfo#ghfo                    |           |                |       |       |           |
| > .8391817                      | At least 1#At least 1        | -.9717479 | .9239607       | -1.05 | 0.293 | -2.782678 |
|                                 | country                      |           |                |       |       |           |
|                                 | Germany                      | 0         | (omitted)      |       |       |           |
|                                 | /phi1_1                      | 1         | (constrained)  |       |       |           |
| > 1.23313                       | /phi1_2                      | 1.009797  | .1139473       | 8.86  | 0.000 | .7864642  |
| > .897637                       | /phi1_3                      | .7338995  | .0835411       | 8.78  | 0.000 | .5701619  |
| > .4977339                      | /phi1_4                      | .3715627  | .0643743       | 5.77  | 0.000 | .2453915  |
|                                 | /phi1_5                      | 0         | (base outcome) |       |       |           |
|                                 | /theta1                      | 16.9221   | 19.70056       | 0.86  | 0.390 | -21.69029 |
| > 55.5345                       | /theta2                      | 18.26493  | 19.63427       | 0.93  | 0.352 | -20.21754 |
| > 56.74741                      | /theta3                      | 15.73383  | 14.38112       | 1.09  | 0.274 | -12.45266 |
| > 43.92031                      | /theta4                      | 9.153959  | 7.505016       | 1.22  | 0.223 | -5.555602 |
| > 23.86352                      | /theta5                      | 0         | (base outcome) |       |       |           |

(sphus=Poorest is the base outcome)  
 results\_impl\_sloCountry\_sup\_1.xls  
 dir : seeout

note: 2.naly omitted because of collinearity.

note: **13.country** omitted because of collinearity.

```
Iteration 0: Log pseudolikelihood = -1191877.3 (not concave)
Iteration 1: Log pseudolikelihood = -810756.71 (not concave)
Iteration 2: Log pseudolikelihood = -550456.95 (not concave)
Iteration 3: Log pseudolikelihood = -380091.27 (not concave)
Iteration 4: Log pseudolikelihood = -234920.03 (not concave)
Iteration 5: Log pseudolikelihood = -171056.04 (not concave)
Iteration 6: Log pseudolikelihood = -154210.24 (not concave)
Iteration 7: Log pseudolikelihood = -139974.08 (not concave)
Iteration 8: Log pseudolikelihood = -137432.65 (not concave)
Iteration 9: Log pseudolikelihood = -129027.41 (not concave)
Iteration 10: Log pseudolikelihood = -126846.01 (not concave)
Iteration 11: Log pseudolikelihood = -125660.39 (not concave)
Iteration 12: Log pseudolikelihood = -125247.39 (not concave)
Iteration 13: Log pseudolikelihood = -124122.29 (not concave)
Iteration 14: Log pseudolikelihood = -122470.5 (not concave)
Iteration 15: Log pseudolikelihood = -121501.63 (not concave)
Iteration 16: Log pseudolikelihood = -119786.52 (not concave)
Iteration 17: Log pseudolikelihood = -118469.27 (not concave)
Iteration 18: Log pseudolikelihood = -117257.42 (not concave)
Iteration 19: Log pseudolikelihood = -116419.02 (not concave)
Iteration 20: Log pseudolikelihood = -115589.18 (not concave)
Iteration 21: Log pseudolikelihood = -114197.49 (not concave)
Iteration 22: Log pseudolikelihood = -113399.52 (not concave)
Iteration 23: Log pseudolikelihood = -113079.16 (not concave)
Iteration 24: Log pseudolikelihood = -112801.41 (not concave)
Iteration 25: Log pseudolikelihood = -112605.06 (not concave)
Iteration 26: Log pseudolikelihood = -112429.44
Iteration 27: Log pseudolikelihood = -111864.87 (not concave)
Iteration 28: Log pseudolikelihood = -111636.5 (not concave)
Iteration 29: Log pseudolikelihood = -111461.87 (not concave)
Iteration 30: Log pseudolikelihood = -111315.37 (not concave)
Iteration 31: Log pseudolikelihood = -111173.03 (not concave)
Iteration 32: Log pseudolikelihood = -111084.14 (not concave)
Iteration 33: Log pseudolikelihood = -111003.51
Iteration 34: Log pseudolikelihood = -110557.13 (not concave)
Iteration 35: Log pseudolikelihood = -110376.34
Iteration 36: Log pseudolikelihood = -110101.8 (not concave)
Iteration 37: Log pseudolikelihood = -109899.04
Iteration 38: Log pseudolikelihood = -109695.93 (not concave)
Iteration 39: Log pseudolikelihood = -109628.11
Iteration 40: Log pseudolikelihood = -109495.52 (not concave)
Iteration 41: Log pseudolikelihood = -109478.5
Iteration 42: Log pseudolikelihood = -109379.62 (not concave)
Iteration 43: Log pseudolikelihood = -109364.86
Iteration 44: Log pseudolikelihood = -109325.15 (not concave)
Iteration 45: Log pseudolikelihood = -109315.11
Iteration 46: Log pseudolikelihood = -109302.4
Iteration 47: Log pseudolikelihood = -109300.91
Iteration 48: Log pseudolikelihood = -109300.89
Iteration 49: Log pseudolikelihood = -109300.89
```

Stereotype logistic regression

Number of obs = **1,202**

Log pseudolikelihood = **-109300.89**

Wald chi2(21) = **294.93**

Prob > chi2 = **0.0000**

( 1) **[phi1\_1]\_cons = 1**

|                   | sphus       | Coefficient      | Robust<br>std. err. | z            | P> z         | [95% con         |
|-------------------|-------------|------------------|---------------------|--------------|--------------|------------------|
| > f. interval]    |             |                  |                     |              |              |                  |
| > <b>1.329976</b> | age         | <b>.3649594</b>  | <b>.4923643</b>     | <b>0.74</b>  | <b>0.459</b> | <b>-.600057</b>  |
| > <b>.0038839</b> | c.age#c.age | <b>-.0024247</b> | <b>.0032187</b>     | <b>-0.75</b> | <b>0.451</b> | <b>-.0087333</b> |
|                   | yedu        | <b>-.1310682</b> | <b>.2389347</b>     | <b>-0.55</b> | <b>0.583</b> | <b>-.5993717</b> |

|   |           |                                      |               |                           |       |       |           |
|---|-----------|--------------------------------------|---------------|---------------------------|-------|-------|-----------|
| > | .3372353  |                                      |               |                           |       |       |           |
| > | .0147657  | c.yedu#c.yedu                        | -.0034455     | .0092916                  | -0.37 | 0.711 | -.0216567 |
| > | .0823201  | thinc                                | -.7818033     | .4408874                  | -1.77 | 0.076 | -1.645927 |
| > | .468251   | c.thinc#c.thinc                      | .0057315      | .2359837                  | 0.02  | 0.981 | -.456788  |
| > | .5510416  | gender<br>Female                     | -.1064907     | .3354818                  | -0.32 | 0.751 | -.764023  |
| > | 1.598697  | household<br>Couple, both responding | .6057132      | .5066336                  | 1.20  | 0.232 | -.3872703 |
| > | 1.682725  | Multiple, at least 1 responding      | .2208954      | .7458453                  | 0.30  | 0.767 | -1.240935 |
| > | 7.407942  | cjs<br>Not applicable                | 4.01063       | 1.733354                  | 2.31  | 0.021 | .6133176  |
| > | 3.244001  | Retired                              | 1.631211      | .8228671                  | 1.98  | 0.047 | .0184212  |
| > | 7.397641  | Unemployed                           | 5.763302      | .8338619                  | 6.91  | 0.000 | 4.128962  |
| > | 12.55023  | Permanently sick or disabled         | 8.670587      | 1.979447                  | 4.38  | 0.000 | 4.790942  |
| > | 3.379242  | Homemaker                            | 1.422181      | .998519                   | 1.42  | 0.154 | -.5348803 |
| > | 5.50328   | Other                                | 2.117094      | 1.727678                  | 1.23  | 0.220 | -1.269092 |
| > | 7.268114  | gali<br>Limited                      | 5.889294      | .7034925                  | 8.37  | 0.000 | 4.510474  |
| > | -.1227184 | lifesat                              | -1.20611      | .5527612                  | -2.18 | 0.029 | -2.289502 |
| > | .0930783  | c.lifesat#c.lifesat                  | .0210395      | .0367552                  | 0.57  | 0.567 | -.0509993 |
| > | -.1447514 | naly<br>At least 1                   | -2.0974       | .9962676                  | -2.11 | 0.035 | -4.050049 |
|   |           | Not applicable                       | 0             | (omitted)                 |       |       |           |
| > | 1.933141  | rhfo<br>At least 1                   | 1.030094      | .4607466                  | 2.24  | 0.025 | .1270475  |
| > | -.3340451 | ghto<br>At least 1                   | -1.241551     | .4630217                  | -2.68 | 0.007 | -2.149057 |
| > | 2.130425  | rhfo#ghto<br>At least 1#At least 1   | .6686901      | .7457969                  | 0.90  | 0.370 | -.7930448 |
|   |           | country<br>Sweden                    | 0             | (omitted)                 |       |       |           |
| > | .9196196  | /phi1_1<br>/phi1_2                   | 1<br>.8360529 | (constrained)<br>.0426368 | 19.61 | 0.000 | .7524863  |
| > | .7237729  | /phi1_3                              | .6311534      | .0472557                  | 13.36 | 0.000 | .5385338  |
| > | .4819599  | /phi1_4                              | .3677214      | .058286                   | 6.31  | 0.000 | .253483   |

|            | /phi1_5 | 0 (base outcome) |          |      |       |           |
|------------|---------|------------------|----------|------|-------|-----------|
| > 44.20924 | /theta1 | 7.35476          | 18.80365 | 0.39 | 0.696 | -29.49972 |
| > 38.65106 | /theta2 | 7.795125         | 15.74312 | 0.50 | 0.620 | -23.06081 |
| > 30.66147 | /theta3 | 7.296293         | 11.92123 | 0.61 | 0.541 | -16.06889 |
| > 18.89185 | /theta4 | 5.157925         | 7.007231 | 0.74 | 0.462 | -8.575996 |
|            | /theta5 | 0 (base outcome) |          |      |       |           |

(sphus=Poor is the base outcome)  
results\_impl\_sloCountry\_sup\_1.xls  
 dir : seeout  
 note: **2.naly** omitted because of collinearity.  
 note: **15.country** omitted because of collinearity.

Iteration 0: Log pseudolikelihood = -312417.04 (not concave)  
 Iteration 1: Log pseudolikelihood = -267922.17 (not concave)  
 Iteration 2: Log pseudolikelihood = -205004.9 (not concave)  
 Iteration 3: Log pseudolikelihood = -165238.51 (not concave)  
 Iteration 4: Log pseudolikelihood = -157292.74 (not concave)  
 Iteration 5: Log pseudolikelihood = -155247.73 (not concave)  
 Iteration 6: Log pseudolikelihood = -153942.95 (not concave)  
 Iteration 7: Log pseudolikelihood = -153411.51 (not concave)  
 Iteration 8: Log pseudolikelihood = -153015.42 (not concave)  
 Iteration 9: Log pseudolikelihood = -152717.52  
 Iteration 10: Log pseudolikelihood = -152688.69 (backed up)  
 Iteration 11: Log pseudolikelihood = -152079.13  
 Iteration 12: Log pseudolikelihood = -152036.42 (not concave)  
 Iteration 13: Log pseudolikelihood = -151996.08  
 Iteration 14: Log pseudolikelihood = -151942.42  
 Iteration 15: Log pseudolikelihood = -151936.16  
 Iteration 16: Log pseudolikelihood = -151923.59  
 Iteration 17: Log pseudolikelihood = -151920.38  
 Iteration 18: Log pseudolikelihood = -151920.16  
 Iteration 19: Log pseudolikelihood = -151920.16

Stereotype logistic regression

Log pseudolikelihood = -151920.16

Number of obs = 1,897  
 Wald chi2(22) = 133.78  
 Prob > chi2 = 0.0000

( 1) [phi1\_1]\_cons = 1

|                | sphus           | Coefficient | Robust<br>std. err. | z     | P> z  | [95% con  |
|----------------|-----------------|-------------|---------------------|-------|-------|-----------|
| > f. interval] |                 |             |                     |       |       |           |
| > 1.317757     | age             | .6526507    | .3393464            | 1.92  | 0.054 | -.012456  |
| > .0005608     | c.age#c.age     | -.0037214   | .0021848            | -1.70 | 0.089 | -.0080035 |
| > .0622842     | yedu            | -.068517    | .0667365            | -1.03 | 0.305 | -.1993182 |
| > .0092787     | c.yedu#c.yedu   | .0022734    | .0035742            | 0.64  | 0.525 | -.0047318 |
| > .4085366     | thinc           | -.2609768   | .3415947            | -0.76 | 0.445 | -.9304901 |
| > .5312212     | c.thinc#c.thinc | .0484114    | .246336             | 0.20  | 0.844 | -.4343983 |
|                | gender          |             |                     |       |       |           |

|   |           |                                 |           |                |       |       |           |
|---|-----------|---------------------------------|-----------|----------------|-------|-------|-----------|
| > | 1.407168  | Female                          | .7802999  | .3198366       | 2.44  | 0.015 | .1534316  |
| > |           | household                       |           |                |       |       |           |
| > | 1.096729  | Couple, both responding         | .5332047  | .2875176       | 1.85  | 0.064 | -.0303194 |
| > | 2.004587  | Multiple, at least 1 responding | .4641868  | .7859328       | 0.59  | 0.555 | -1.076213 |
| > |           | cjs                             |           |                |       |       |           |
| > | 5.564626  | Not applicable                  | 2.412267  | 1.608376       | 1.50  | 0.134 | -.7400921 |
| > | 2.574062  | Retired                         | 1.013027  | .796461        | 1.27  | 0.203 | -.5480082 |
| > | 4.962921  | Unemployed                      | 2.065201  | 1.478455       | 1.40  | 0.162 | -.8325177 |
| > | 4.5484    | Permanently sick or disabled    | 2.582481  | 1.003038       | 2.57  | 0.010 | .6165622  |
| > | 3.005004  | Homemaker                       | 1.363767  | .8373809       | 1.63  | 0.103 | -.277469  |
| > | 3.007539  | Other                           | 1.069725  | .9886987       | 1.08  | 0.279 | -.8680892 |
| > |           | gali                            |           |                |       |       |           |
| > | 7.3057    | Limited                         | 5.891997  | .72129         | 8.17  | 0.000 | 4.478295  |
| > | .3475262  | lifesat                         | -.2478803 | .3037844       | -0.82 | 0.415 | -.8432869 |
| > | .0197233  | c.lifesat#c.lifesat             | -.0228082 | .0217002       | -1.05 | 0.293 | -.0653398 |
| > |           | naly                            |           |                |       |       |           |
| > | -.1766243 | At least 1                      | -.6861278 | .2599555       | -2.64 | 0.008 | -1.195631 |
| > |           | Not applicable                  | 0         | (omitted)      |       |       |           |
| > |           | rhfo                            |           |                |       |       |           |
| > | 2.320258  | At least 1                      | 1.553099  | .3914144       | 3.97  | 0.000 | .7859413  |
| > |           | ghfo                            |           |                |       |       |           |
| > | 1.093566  | At least 1                      | -.0682862 | .5927927       | -0.12 | 0.908 | -1.230139 |
| > |           | rhfo#ghfo                       |           |                |       |       |           |
| > | 4.923937  | At least 1#At least 1           | 1.877129  | 1.554522       | 1.21  | 0.227 | -1.169679 |
| > |           | country                         |           |                |       |       |           |
| > |           | Spain                           | 0         | (omitted)      |       |       |           |
| > |           | /phi1_1                         | 1         | (constrained)  |       |       |           |
| > | 1.038271  | /phi1_2                         | .8394212  | .1014561       | 8.27  | 0.000 | .6405709  |
| > | .7242207  | /phi1_3                         | .5914926  | .0677197       | 8.73  | 0.000 | .4587645  |
| > | .3935304  | /phi1_4                         | .312697   | .0412423       | 7.58  | 0.000 | .2318636  |
| > |           | /phi1_5                         | 0         | (base outcome) |       |       |           |
| > |           | /theta1                         | 26.4022   | 12.86557       | 2.05  | 0.040 | 1.186137  |
| > | 51.61826  | /theta2                         | 23.9154   | 10.2819        | 2.33  | 0.020 | 3.763242  |
| > | 44.06756  | /theta3                         | 18.59699  | 7.322376       | 2.54  | 0.011 | 4.245394  |
| > | 32.94858  | /theta4                         | 10.53545  | 3.942982       | 2.67  | 0.008 | 2.807347  |
| > | 18.26355  | /theta5                         | 0         | (base outcome) |       |       |           |

(sphus=poor is the base outcome)  
 results\_impl\_sloCountry\_sup\_1.xls  
 dir : seeout  
 note: **2.naly** omitted because of collinearity.  
 note: **16.country** omitted because of collinearity.

```
Iteration 0: Log pseudolikelihood = -787868.04 (not concave)
Iteration 1: Log pseudolikelihood = -570109.49 (not concave)
Iteration 2: Log pseudolikelihood = -352226.66 (not concave)
Iteration 3: Log pseudolikelihood = -235226.43 (not concave)
Iteration 4: Log pseudolikelihood = -191732.64 (not concave)
Iteration 5: Log pseudolikelihood = -168259.17 (not concave)
Iteration 6: Log pseudolikelihood = -157811.81 (not concave)
Iteration 7: Log pseudolikelihood = -154271.03 (not concave)
Iteration 8: Log pseudolikelihood = -152486.66 (not concave)
Iteration 9: Log pseudolikelihood = -150722.55 (not concave)
Iteration 10: Log pseudolikelihood = -149852.11 (not concave)
Iteration 11: Log pseudolikelihood = -148408.65 (not concave)
Iteration 12: Log pseudolikelihood = -147765.06 (not concave)
Iteration 13: Log pseudolikelihood = -146365.76 (not concave)
Iteration 14: Log pseudolikelihood = -146216.37 (not concave)
Iteration 15: Log pseudolikelihood = -146103.48 (not concave)
Iteration 16: Log pseudolikelihood = -146007.2 (not concave)
Iteration 17: Log pseudolikelihood = -145919.7 (not concave)
Iteration 18: Log pseudolikelihood = -145846.11 (not concave)
Iteration 19: Log pseudolikelihood = -145774.59 (not concave)
Iteration 20: Log pseudolikelihood = -145708.73 (not concave)
Iteration 21: Log pseudolikelihood = -145645.37 (not concave)
Iteration 22: Log pseudolikelihood = -145583.13 (not concave)
Iteration 23: Log pseudolikelihood = -145523.23 (not concave)
Iteration 24: Log pseudolikelihood = -145465.41 (not concave)
Iteration 25: Log pseudolikelihood = -145409.12 (not concave)
Iteration 26: Log pseudolikelihood = -145354.18 (not concave)
Iteration 27: Log pseudolikelihood = -145300.82 (not concave)
Iteration 28: Log pseudolikelihood = -145248.8 (not concave)
Iteration 29: Log pseudolikelihood = -145198.09 (not concave)
Iteration 30: Log pseudolikelihood = -145148.56 (not concave)
Iteration 31: Log pseudolikelihood = -145100.3 (not concave)
Iteration 32: Log pseudolikelihood = -145053.15 (not concave)
Iteration 33: Log pseudolikelihood = -145007.18 (not concave)
Iteration 34: Log pseudolikelihood = -144423.41 (not concave)
Iteration 35: Log pseudolikelihood = -144351.57 (not concave)
Iteration 36: Log pseudolikelihood = -144301.04 (not concave)
Iteration 37: Log pseudolikelihood = -143882.72 (not concave)
Iteration 38: Log pseudolikelihood = -143811.38 (not concave)
Iteration 39: Log pseudolikelihood = -143635.24 (not concave)
Iteration 40: Log pseudolikelihood = -143524.78 (not concave)
Iteration 41: Log pseudolikelihood = -143458.06 (not concave)
Iteration 42: Log pseudolikelihood = -143450.55 (not concave)
Iteration 43: Log pseudolikelihood = -143446.95 (not concave)
Iteration 44: Log pseudolikelihood = -143446.87 (not concave)
Iteration 45: Log pseudolikelihood = -143446.87 (not concave)
```

Stereotype logistic regression

Number of obs = **1,766**  
 Wald chi2(22) = **133.67**  
 Prob > chi2 = **0.0000**

Log pseudolikelihood = **-143446.87**

( 1) [phi1\_1]\_cons = 1

|                | sphus       | Coefficient | Robust<br>std. err. | z     | P> z  | [95% con  |
|----------------|-------------|-------------|---------------------|-------|-------|-----------|
| > f. interval] |             |             |                     |       |       |           |
| > 1.055351     | age         | .3252495    | .3725075            | 0.87  | 0.383 | -.4048518 |
| > .0030323     | c.age#c.age | -.0017784   | .0024545            | -0.72 | 0.469 | -.0065892 |

|   |           |                                      |            |           |       |       |           |
|---|-----------|--------------------------------------|------------|-----------|-------|-------|-----------|
| > | .0347389  | yedu                                 | -1.1871652 | .1132184  | -1.65 | 0.098 | -.4090693 |
| > | .0165683  | c.yedu#c.yedu                        | .0058854   | .0054505  | 1.08  | 0.280 | -.0047975 |
| > | .4890176  | thinc                                | -.1205965  | .3110333  | -0.39 | 0.698 | -.7302106 |
| > | .4000477  | c.thinc#c.thinc                      | .0123462   | .1978105  | 0.06  | 0.950 | -.3753553 |
| > | .7546436  | gender<br>Female                     | .2338652   | .2657082  | 0.88  | 0.379 | -.2869132 |
| > | .9458505  | household<br>Couple, both responding | .3078329   | .3255252  | 0.95  | 0.344 | -.3301848 |
| > | 1.831846  | Multiple, at least 1 responding      | .6531141   | .6014048  | 1.09  | 0.277 | -.5256178 |
| > | 11.26243  | cjs<br>Not applicable                | 2.429669   | 4.506592  | 0.54  | 0.590 | -6.403089 |
| > | 4.457242  | Retired                              | 2.452065   | 1.023068  | 2.40  | 0.017 | .446888   |
| > | 4.123566  | Unemployed                           | .5168234   | 1.840208  | 0.28  | 0.779 | -3.089919 |
| > | 7.934572  | Permanently sick or disabled         | 5.027164   | 1.483399  | 3.39  | 0.001 | 2.119756  |
| > | 4.706671  | Homemaker                            | 2.600885   | 1.0744    | 2.42  | 0.015 | .4950987  |
| > | 5.656681  | Other                                | 3.41168    | 1.14543   | 2.98  | 0.003 | 1.166678  |
| > | 6.154729  | gali<br>Limited                      | 5.04295    | .5672447  | 8.89  | 0.000 | 3.93117   |
| > | -.3118366 | lifesat                              | -.8303756  | .2645656  | -3.14 | 0.002 | -1.348915 |
| > | .0578004  | c.lifesat#c.lifesat                  | .0164191   | .0211133  | 0.78  | 0.437 | -.0249622 |
| > | .566452   | naly<br>At least 1                   | .0647908   | .2559543  | 0.25  | 0.800 | -.4368704 |
|   |           | Not applicable                       | 0          | (omitted) |       |       |           |
| > | 1.914272  | rhfo<br>At least 1                   | 1.256519   | .3355943  | 3.74  | 0.000 | .5987663  |
| > | .6283523  | ghfo<br>At least 1                   | -.1651524  | .4048568  | -0.41 | 0.683 | -.9586571 |
| > | .6592304  | rhfo#ghfo<br>At least 1#At least 1   | -1.021261  | .8574092  | -1.19 | 0.234 | -2.701752 |
|   |           | country<br>Italy                     | 0          | (omitted) |       |       |           |
| > | 1.052641  | /phi1_1<br>/phi1_2                   | .871273    | .0925363  | 9.42  | 0.000 | .6899052  |
| > | .8733099  | /phi1_3                              | .7278253   | .0742282  | 9.81  | 0.000 | .5823408  |

|   |          |         |          |                |      |       |           |
|---|----------|---------|----------|----------------|------|-------|-----------|
| > | .506879  | /phi1_4 | .4041525 | .0524125       | 7.71 | 0.000 | .3014259  |
|   |          | /phi1_5 | 0        | (base outcome) |      |       |           |
| > | 40.25684 | /theta1 | 12.92188 | 13.94666       | 0.93 | 0.354 | -14.41309 |
| > | 36.09089 | /theta2 | 12.29707 | 12.13993       | 1.01 | 0.311 | -11.49676 |
| > | 31.7622  | /theta3 | 11.91965 | 10.12393       | 1.18 | 0.239 | -7.922895 |
| > | 18.88783 | /theta4 | 7.790044 | 5.662242       | 1.38 | 0.169 | -3.307746 |
|   |          | /theta5 | 0        | (base outcome) |      |       |           |

(sphus=Poor is the base outcome)

results\_impl\_sloCountry\_sup\_1.xls

dir : seeout

note: **2.naly** omitted because of collinearity.

note: **17.country** omitted because of collinearity.

Iteration 0: Log pseudolikelihood = -667534.89 (not concave)  
 Iteration 1: Log pseudolikelihood = -459215.32 (not concave)  
 Iteration 2: Log pseudolikelihood = -248669.56 (not concave)  
 Iteration 3: Log pseudolikelihood = -194717.55 (not concave)  
 Iteration 4: Log pseudolikelihood = -183818.36 (not concave)  
 Iteration 5: Log pseudolikelihood = -177829.63 (not concave)  
 Iteration 6: Log pseudolikelihood = -176987.72 (not concave)  
 Iteration 7: Log pseudolikelihood = -176142.89 (not concave)  
 Iteration 8: Log pseudolikelihood = -175581.32 (not concave)  
 Iteration 9: Log pseudolikelihood = -175102.4 (not concave)  
 Iteration 10: Log pseudolikelihood = -174294.37 (not concave)  
 Iteration 11: Log pseudolikelihood = -173494.19 (not concave)  
 Iteration 12: Log pseudolikelihood = -173036.68 (not concave)  
 Iteration 13: Log pseudolikelihood = -172625.3 (not concave)  
 Iteration 14: Log pseudolikelihood = -172226.72 (not concave)  
 Iteration 15: Log pseudolikelihood = -172130.08 (not concave)  
 Iteration 16: Log pseudolikelihood = -171953.44 (not concave)  
 Iteration 17: Log pseudolikelihood = -171802.54 (not concave)  
 Iteration 18: Log pseudolikelihood = -171726.58 (not concave)  
 Iteration 19: Log pseudolikelihood = -171695.43 (not concave)  
 Iteration 20: Log pseudolikelihood = -171681.18 (not concave)  
 Iteration 21: Log pseudolikelihood = -171679.01 (not concave)  
 Iteration 22: Log pseudolikelihood = -171678.98 (not concave)  
 Iteration 23: Log pseudolikelihood = -171678.98 (not concave)

Stereotype logistic regression

Number of obs = 2,103

Wald chi2(22) = 366.57

Prob > chi2 = 0.0000

Log pseudolikelihood = -171678.98

( 1) [phi1\_1]\_cons = 1

|                | sphus         | Coefficient | Robust<br>std. err. | z     | P> z  | [95% con  |
|----------------|---------------|-------------|---------------------|-------|-------|-----------|
| > f. interval] |               |             |                     |       |       |           |
| >              | age           | .1857223    | .2911044            | 0.64  | 0.523 | -.3848318 |
| >              | c.age#c.age   | -.0008943   | .0019077            | -0.47 | 0.639 | -.0046333 |
| >              | yedu          | -.1768166   | .1260065            | -1.40 | 0.161 | -.4237849 |
| >              | c.yedu#c.yedu | .0031153    | .0050226            | 0.62  | 0.535 | -.0067289 |
| >              | thinc         | -1.002294   | .2757532            | -3.63 | 0.000 | -1.54276  |

|   |            |                                 |           |                |       |       |            |
|---|------------|---------------------------------|-----------|----------------|-------|-------|------------|
| > | - .4618273 |                                 |           |                |       |       |            |
| > | .3122722   | c.thinc#c.thinc                 | - .024317 | .1717323       | -0.14 | 0.887 | - .3609061 |
|   |            | gender                          |           |                |       |       |            |
| > | .468346    | Female                          | .012283   | .2326895       | 0.05  | 0.958 | - .4437801 |
|   |            | household                       |           |                |       |       |            |
|   |            | Couple, both responding         | .6128879  | .3050258       | 2.01  | 0.045 | .0150483   |
| > | 1.210728   | Multiple, at least 1 responding | 1.861639  | .5006421       | 3.72  | 0.000 | .8803989   |
| > | 2.84288    |                                 |           |                |       |       |            |
|   |            | cjs                             |           |                |       |       |            |
|   |            | Not applicable                  | 1.179655  | 1.41189        | 0.84  | 0.403 | -1.587598  |
| > | 3.946907   |                                 |           |                |       |       |            |
|   |            | Retired                         | .5304155  | 1.051448       | 0.50  | 0.614 | -1.530384  |
| > | 2.591215   |                                 |           |                |       |       |            |
|   |            | Unemployed                      | -5.751171 | 1.350747       | -4.26 | 0.000 | -8.398586  |
| > | -3.103755  |                                 |           |                |       |       |            |
|   |            | Permanently sick or disabled    | 6.713779  | 3.283366       | 2.04  | 0.041 | .2784995   |
| > | 13.14906   |                                 |           |                |       |       |            |
|   |            | Homemaker                       | .8531126  | 1.190705       | 0.72  | 0.474 | -1.480626  |
| > | 3.186851   |                                 |           |                |       |       |            |
|   |            | Other                           | -2.784169 | 1.586071       | -1.76 | 0.079 | -5.89281   |
| > | .3244732   |                                 |           |                |       |       |            |
|   |            | gali                            |           |                |       |       |            |
|   |            | Limited                         | 5.031066  | .413778        | 12.16 | 0.000 | 4.220075   |
| > | 5.842056   |                                 |           |                |       |       |            |
|   |            | lifesat                         | -.915691  | .3206211       | -2.86 | 0.004 | -1.544097  |
| > | -.2872852  |                                 |           |                |       |       |            |
|   |            | c.lifesat#c.lifesat             | .0083343  | .024152        | 0.35  | 0.730 | -.0390028  |
| > | .0556714   |                                 |           |                |       |       |            |
|   |            | naly                            |           |                |       |       |            |
|   |            | At least 1                      | -.4852201 | .3458125       | -1.40 | 0.161 | -1.163     |
| > | .1925598   |                                 |           |                |       |       |            |
|   |            | Not applicable                  | 0         | (omitted)      |       |       |            |
|   |            | rhfo                            |           |                |       |       |            |
|   |            | At least 1                      | 1.325555  | .3420693       | 3.88  | 0.000 | .6551116   |
| > | 1.995999   |                                 |           |                |       |       |            |
|   |            | ghfo                            |           |                |       |       |            |
|   |            | At least 1                      | -.9544238 | .2984068       | -3.20 | 0.001 | -1.53929   |
| > | -.3695572  |                                 |           |                |       |       |            |
|   |            | rhfo#ghfo                       |           |                |       |       |            |
|   |            | At least 1#At least 1           | -.6675479 | .5827259       | -1.15 | 0.252 | -1.80967   |
| > | .4745738   |                                 |           |                |       |       |            |
|   |            | country                         |           |                |       |       |            |
|   |            | France                          | 0         | (omitted)      |       |       |            |
|   |            | /phil_1                         | 1         | (constrained)  |       |       |            |
|   |            | /phil_2                         | .8369695  | .0587184       | 14.25 | 0.000 | .7218836   |
| > | .9520554   |                                 |           |                |       |       |            |
|   |            | /phil_3                         | .6011685  | .0404529       | 14.86 | 0.000 | .5218823   |
| > | .6804546   |                                 |           |                |       |       |            |
|   |            | /phil_4                         | .3096544  | .0333741       | 9.28  | 0.000 | .2442423   |
| > | .3750665   |                                 |           |                |       |       |            |
|   |            | /phil_5                         | 0         | (base outcome) |       |       |            |
|   |            | /theta1                         | 2.937853  | 11.24556       | 0.26  | 0.794 | -19.10303  |
| > | 24.97874   |                                 |           |                |       |       |            |
|   |            | /theta2                         | 3.755714  | 9.42354        | 0.40  | 0.690 | -14.71408  |

|   |          |         |          |                |      |       |           |
|---|----------|---------|----------|----------------|------|-------|-----------|
| > | 22.22551 | /theta3 | 4.672758 | 6.788693       | 0.69 | 0.491 | -8.632836 |
| > | 17.97835 | /theta4 | 3.026864 | 3.532908       | 0.86 | 0.392 | -3.897507 |
| > | 9.951236 | /theta5 | 0        | (base outcome) |      |       |           |

(sphus=Poor is the base outcome)

results\_impl\_sloCountry\_sup\_1.xls

dir : seeout

note: **2.naly** omitted because of collinearity.

note: **18.country** omitted because of collinearity.

```

Iteration 0: Log pseudolikelihood = -622292.44 (not concave)
Iteration 1: Log pseudolikelihood = -401855.62 (not concave)
Iteration 2: Log pseudolikelihood = -170592.11 (not concave)
Iteration 3: Log pseudolikelihood = -124207.76 (not concave)
Iteration 4: Log pseudolikelihood = -101965.38 (not concave)
Iteration 5: Log pseudolikelihood = -99813.234 (not concave)
Iteration 6: Log pseudolikelihood = -99207.454 (not concave)
Iteration 7: Log pseudolikelihood = -98875.34 (not concave)
Iteration 8: Log pseudolikelihood = -98535.952 (not concave)
Iteration 9: Log pseudolikelihood = -98409.205 (not concave)
Iteration 10: Log pseudolikelihood = -98333.555 (not concave)
Iteration 11: Log pseudolikelihood = -98260.673 (not concave)
Iteration 12: Log pseudolikelihood = -98106.059 (not concave)
Iteration 13: Log pseudolikelihood = -98038.499 (not concave)
Iteration 14: Log pseudolikelihood = -98029.72 (not concave)
Iteration 15: Log pseudolikelihood = -98023.689 (not concave)
Iteration 16: Log pseudolikelihood = -98017.854 (not concave)
Iteration 17: Log pseudolikelihood = -97998.538 (not concave)
Iteration 18: Log pseudolikelihood = -97982.705 (not concave)
Iteration 19: Log pseudolikelihood = -97957.164 (not concave)
Iteration 20: Log pseudolikelihood = -97933.518 (not concave)
Iteration 21: Log pseudolikelihood = -97925.281 (not concave)
Iteration 22: Log pseudolikelihood = -97916.772 (not concave)
Iteration 23: Log pseudolikelihood = -97900.768 (not concave)
Iteration 24: Log pseudolikelihood = -97889.1 (not concave)
Iteration 25: Log pseudolikelihood = -97883.338 (not concave)
Iteration 26: Log pseudolikelihood = -97877.531 (not concave)
Iteration 27: Log pseudolikelihood = -97872.154 (not concave)
Iteration 28: Log pseudolikelihood = -97866.849 (not concave)
Iteration 29: Log pseudolikelihood = -97861.401 (not concave)
Iteration 30: Log pseudolikelihood = -97855.886 (not concave)
Iteration 31: Log pseudolikelihood = -97850.384 (not concave)
Iteration 32: Log pseudolikelihood = -97844.822 (not concave)
Iteration 33: Log pseudolikelihood = -97839.197 (not concave)
Iteration 34: Log pseudolikelihood = -97833.511 (not concave)
Iteration 35: Log pseudolikelihood = -97827.79 (not concave)
Iteration 36: Log pseudolikelihood = -97822.002 (not concave)
Iteration 37: Log pseudolikelihood = -97816.166 (not concave)
Iteration 38: Log pseudolikelihood = -97810.261 (not concave)
Iteration 39: Log pseudolikelihood = -97804.311 (not concave)
Iteration 40: Log pseudolikelihood = -97798.288 (not concave)
Iteration 41: Log pseudolikelihood = -97792.215 (not concave)
Iteration 42: Log pseudolikelihood = -97786.068 (not concave)
Iteration 43: Log pseudolikelihood = -97779.869 (not concave)
Iteration 44: Log pseudolikelihood = -97773.592 (not concave)
Iteration 45: Log pseudolikelihood = -97767.261 (not concave)
Iteration 46: Log pseudolikelihood = -97760.848 (not concave)
Iteration 47: Log pseudolikelihood = -97754.379 (not concave)
Iteration 48: Log pseudolikelihood = -97747.824 (not concave)
Iteration 49: Log pseudolikelihood = -97741.209 (not concave)
Iteration 50: Log pseudolikelihood = -97734.506 (not concave)
Iteration 51: Log pseudolikelihood = -97727.74 (not concave)
Iteration 52: Log pseudolikelihood = -97720.881 (not concave)
Iteration 53: Log pseudolikelihood = -97713.956 (not concave)
Iteration 54: Log pseudolikelihood = -97706.934 (not concave)
Iteration 55: Log pseudolikelihood = -97699.843 (not concave)
Iteration 56: Log pseudolikelihood = -97692.649 (not concave)
Iteration 57: Log pseudolikelihood = -97685.383 (not concave)

```

Iteration 58: Log pseudolikelihood = -97678.011 (not concave)  
Iteration 59: Log pseudolikelihood = -97670.561 (not concave)  
Iteration 60: Log pseudolikelihood = -97663.001 (not concave)  
Iteration 61: Log pseudolikelihood = -97655.359 (not concave)  
Iteration 62: Log pseudolikelihood = -97647.6 (not concave)  
Iteration 63: Log pseudolikelihood = -97639.756 (not concave)  
Iteration 64: Log pseudolikelihood = -97631.789 (not concave)  
Iteration 65: Log pseudolikelihood = -97623.733 (not concave)  
Iteration 66: Log pseudolikelihood = -97615.547 (not concave)  
Iteration 67: Log pseudolikelihood = -97607.267 (not concave)  
Iteration 68: Log pseudolikelihood = -97598.851 (not concave)  
Iteration 69: Log pseudolikelihood = -97590.336 (not concave)  
Iteration 70: Log pseudolikelihood = -97581.678 (not concave)  
Iteration 71: Log pseudolikelihood = -97572.914 (not concave)  
Iteration 72: Log pseudolikelihood = -97564.001 (not concave)  
Iteration 73: Log pseudolikelihood = -97554.976 (not concave)  
Iteration 74: Log pseudolikelihood = -97545.793 (not concave)  
Iteration 75: Log pseudolikelihood = -97536.493 (not concave)  
Iteration 76: Log pseudolikelihood = -97527.027 (not concave)  
Iteration 77: Log pseudolikelihood = -97517.435 (not concave)  
Iteration 78: Log pseudolikelihood = -97507.67 (not concave)  
Iteration 79: Log pseudolikelihood = -97497.772 (not concave)  
Iteration 80: Log pseudolikelihood = -97487.69 (not concave)  
Iteration 81: Log pseudolikelihood = -97477.468 (not concave)  
Iteration 82: Log pseudolikelihood = -97467.052 (not concave)  
Iteration 83: Log pseudolikelihood = -97456.489 (not concave)  
Iteration 84: Log pseudolikelihood = -97445.72 (not concave)  
Iteration 85: Log pseudolikelihood = -97434.796 (not concave)  
Iteration 86: Log pseudolikelihood = -97423.655 (not concave)  
Iteration 87: Log pseudolikelihood = -97412.349 (not concave)  
Iteration 88: Log pseudolikelihood = -97400.814 (not concave)  
Iteration 89: Log pseudolikelihood = -97389.105 (not concave)  
Iteration 90: Log pseudolikelihood = -97377.155 (not concave)  
Iteration 91: Log pseudolikelihood = -97365.021 (not concave)  
Iteration 92: Log pseudolikelihood = -97352.632 (not concave)  
Iteration 93: Log pseudolikelihood = -97340.049 (not concave)  
Iteration 94: Log pseudolikelihood = -97327.198 (not concave)  
Iteration 95: Log pseudolikelihood = -97314.141 (not concave)  
Iteration 96: Log pseudolikelihood = -97300.802 (not concave)  
Iteration 97: Log pseudolikelihood = -97287.246 (not concave)  
Iteration 98: Log pseudolikelihood = -97273.393 (not concave)  
Iteration 99: Log pseudolikelihood = -97259.313 (not concave)  
Iteration 100: Log pseudolikelihood = -97244.92 (not concave)  
Iteration 101: Log pseudolikelihood = -97230.289 (not concave)  
Iteration 102: Log pseudolikelihood = -97215.331 (not concave)  
Iteration 103: Log pseudolikelihood = -97200.123 (not concave)  
Iteration 104: Log pseudolikelihood = -97184.573 (not concave)  
Iteration 105: Log pseudolikelihood = -97168.764 (not concave)  
Iteration 106: Log pseudolikelihood = -97152.598 (not concave)  
Iteration 107: Log pseudolikelihood = -97136.163 (not concave)  
Iteration 108: Log pseudolikelihood = -97119.358 (not concave)  
Iteration 109: Log pseudolikelihood = -97102.276 (not concave)  
Iteration 110: Log pseudolikelihood = -97084.813 (not concave)  
Iteration 111: Log pseudolikelihood = -97067.068 (not concave)  
Iteration 112: Log pseudolikelihood = -97048.932 (not concave)  
Iteration 113: Log pseudolikelihood = -97030.511 (not concave)  
Iteration 114: Log pseudolikelihood = -97011.693 (not concave)  
Iteration 115: Log pseudolikelihood = -96992.592 (not concave)  
Iteration 116: Log pseudolikelihood = -96973.091 (not concave)  
Iteration 117: Log pseudolikelihood = -96953.313 (not concave)  
Iteration 118: Log pseudolikelihood = -96933.14 (not concave)  
Iteration 119: Log pseudolikelihood = -96912.702 (not concave)  
Iteration 120: Log pseudolikelihood = -96891.879 (not concave)  
Iteration 121: Log pseudolikelihood = -96870.811 (not concave)  
Iteration 122: Log pseudolikelihood = -96849.377 (not concave)  
Iteration 123: Log pseudolikelihood = -96827.726 (not concave)  
Iteration 124: Log pseudolikelihood = -96805.737 (not concave)  
Iteration 125: Log pseudolikelihood = -96783.569 (not concave)  
Iteration 126: Log pseudolikelihood = -96761.104 (not concave)  
Iteration 127: Log pseudolikelihood = -96738.507 (not concave)  
Iteration 128: Log pseudolikelihood = -96715.664 (not concave)  
Iteration 129: Log pseudolikelihood = -96692.749 (not concave)

```

Iteration 130: Log pseudolikelihood = -96669.651 (not concave)
Iteration 131: Log pseudolikelihood = -96646.55 (not concave)
Iteration 132: Log pseudolikelihood = -96623.341 (not concave)
Iteration 133: Log pseudolikelihood = -96600.209 (not concave)
Iteration 134: Log pseudolikelihood = -96577.052 (not concave)
Iteration 135: Log pseudolikelihood = -96554.059 (not concave)
Iteration 136: Log pseudolikelihood = -96531.134 (not concave)
Iteration 137: Log pseudolikelihood = -96508.463 (not concave)
Iteration 138: Log pseudolikelihood = -96485.956 (not concave)
Iteration 139: Log pseudolikelihood = -96463.795 (not concave)
Iteration 140: Log pseudolikelihood = -96441.894 (not concave)
Iteration 141: Log pseudolikelihood = -96420.426 (not concave)
Iteration 142: Log pseudolikelihood = -96399.31 (not concave)
Iteration 143: Log pseudolikelihood = -96378.707 (not concave)
Iteration 144: Log pseudolikelihood = -96358.538 (not concave)
Iteration 145: Log pseudolikelihood = -96338.951 (not concave)
Iteration 146: Log pseudolikelihood = -96319.868 (not concave)
Iteration 147: Log pseudolikelihood = -96301.42 (not concave)
Iteration 148: Log pseudolikelihood = -96145.857 (not concave)
Iteration 149: Log pseudolikelihood = -96011.794 (not concave)
Iteration 150: Log pseudolikelihood = -95893.197 (not concave)
Iteration 151: Log pseudolikelihood = -95872.941 (not concave)
Iteration 152: Log pseudolikelihood = -95846.025 (not concave)
Iteration 153: Log pseudolikelihood = -95828.043 (not concave)
Iteration 154: Log pseudolikelihood = -95824.69 (not concave)
Iteration 155: Log pseudolikelihood = -95823.28 (not concave)
Iteration 156: Log pseudolikelihood = -95823.275 (not concave)

```

Stereotype logistic regression

Number of obs = 1,040

Log pseudolikelihood = -95823.275

Wald chi2(21) = 200.99

Prob > chi2 = 0.0000

( 1) [phi1\_1]\_cons = 1

|                |          | sphus                                | Coefficient | Robust<br>std. err. | z     | P> z  | [95% con  |
|----------------|----------|--------------------------------------|-------------|---------------------|-------|-------|-----------|
| > f. interval] |          |                                      |             |                     |       |       |           |
| >              | 1.297007 | age                                  | .333744     | .4914695            | 0.68  | 0.497 | -.6295186 |
| >              | .0042366 | c.age#c.age                          | -.0020372   | .003201             | -0.64 | 0.524 | -.0083111 |
| >              | .875233  | yedu                                 | .2123325    | .3382208            | 0.63  | 0.530 | -.450568  |
| >              | .0127844 | c.yedu#c.yedu                        | -.0146904   | .014018             | -1.05 | 0.295 | -.0421653 |
| >              | .9083674 | thinc                                | -.0412414   | .4845032            | -0.09 | 0.932 | -.9908502 |
| >              | .1780792 | c.thinc#c.thinc                      | -.3743848   | .2818746            | -1.33 | 0.184 | -.9268489 |
| >              | .5245766 | gender<br>Female                     | -.133385    | .3357009            | -0.40 | 0.691 | -.7913467 |
| >              | 1.718575 | household<br>Couple, both responding | .8022899    | .4675011            | 1.72  | 0.086 | -.1139954 |
| >              | 1.802623 | Multiple, at least 1 responding      | .5551484    | .6364781            | 0.87  | 0.383 | -.6923258 |
| >              | 3.949129 | cjs<br>Not applicable                | 1.108238    | 1.449461            | 0.76  | 0.445 | -1.732653 |
|                |          | Retired                              | .9844935    | .7998759            | 1.23  | 0.218 | -.5832345 |

|       |           |                              |           |                |       |       |           |
|-------|-----------|------------------------------|-----------|----------------|-------|-------|-----------|
| >     | 2.552221  |                              |           |                |       |       |           |
|       |           | Unemployed                   | 2.095819  | 1.445402       | 1.45  | 0.147 | -.7371165 |
| >     | 4.928754  |                              |           |                |       |       |           |
|       |           | Permanently sick or disabled | 2.181045  | 2.226929       | 0.98  | 0.327 | -2.183655 |
| >     | 6.545746  |                              |           |                |       |       |           |
|       |           | Other                        | -.7417575 | 1.73318        | -0.43 | 0.669 | -4.138728 |
| >     | 2.655213  |                              |           |                |       |       |           |
|       |           | gali                         |           |                |       |       |           |
|       |           | Limited                      | 5.784298  | .6362618       | 9.09  | 0.000 | 4.537248  |
| >     | 7.031348  |                              |           |                |       |       |           |
|       |           | lifesat                      | -.8964498 | .664147        | -1.35 | 0.177 | -2.198154 |
| >     | .4052545  |                              |           |                |       |       |           |
|       |           | c.lifesat#c.lifesat          | -.0090869 | .0446989       | -0.20 | 0.839 | -.0966951 |
| >     | .0785213  |                              |           |                |       |       |           |
|       |           | naly                         |           |                |       |       |           |
|       |           | At least 1                   | -1.784635 | .7060062       | -2.53 | 0.011 | -3.168382 |
| >     | -.4008882 |                              |           |                |       |       |           |
|       |           | Not applicable               | 0         | (omitted)      |       |       |           |
|       |           | rhfo                         |           |                |       |       |           |
|       |           | At least 1                   | 2.051567  | .4693396       | 4.37  | 0.000 | 1.131679  |
| >     | 2.971456  |                              |           |                |       |       |           |
|       |           | ghfo                         |           |                |       |       |           |
|       |           | At least 1                   | .3094082  | .436208        | 0.71  | 0.478 | -.5455437 |
| >     | 1.16436   |                              |           |                |       |       |           |
|       |           | rhfo#ghfo                    |           |                |       |       |           |
|       |           | At least 1#At least 1        | -1.209668 | .708322        | -1.71 | 0.088 | -2.597953 |
| >     | .1786179  |                              |           |                |       |       |           |
|       |           | country                      |           |                |       |       |           |
|       |           | Denmark                      | 0         | (omitted)      |       |       |           |
| <hr/> |           |                              |           |                |       |       |           |
|       |           | /phi1_1                      | 1         | (constrained)  |       |       |           |
|       |           | /phi1_2                      | .7736706  | .04944         | 15.65 | 0.000 | .67677    |
| >     | .8705712  |                              |           |                |       |       |           |
|       |           | /phi1_3                      | .6442271  | .0490936       | 13.12 | 0.000 | .5480054  |
| >     | .7404489  |                              |           |                |       |       |           |
|       |           | /phi1_4                      | .3406229  | .0621325       | 5.48  | 0.000 | .2188455  |
| >     | .4624003  |                              |           |                |       |       |           |
|       |           | /phi1_5                      | 0         | (base outcome) |       |       |           |
| <hr/> |           |                              |           |                |       |       |           |
|       |           | /theta1                      | 9.883957  | 19.02633       | 0.52  | 0.603 | -27.40697 |
| >     | 47.17488  |                              |           |                |       |       |           |
|       |           | /theta2                      | 9.552478  | 14.69627       | 0.65  | 0.516 | -19.25168 |
| >     | 38.35664  |                              |           |                |       |       |           |
|       |           | /theta3                      | 8.442579  | 12.28251       | 0.69  | 0.492 | -15.63069 |
| >     | 32.51585  |                              |           |                |       |       |           |
|       |           | /theta4                      | 5.348213  | 6.550781       | 0.82  | 0.414 | -7.491081 |
| >     | 18.18751  |                              |           |                |       |       |           |
|       |           | /theta5                      | 0         | (base outcome) |       |       |           |

(sphus=Poor is the base outcome)  
results\_impl\_sloCountry\_sup\_1.xls  
dir : seeout  
note: **2.naly** omitted because of collinearity.  
note: **20.country** omitted because of collinearity.

```

Iteration 0: Log pseudolikelihood = -1052433.7 (not concave)
Iteration 1: Log pseudolikelihood = -279640.58 (not concave)
Iteration 2: Log pseudolikelihood = -225537.41 (not concave)
Iteration 3: Log pseudolikelihood = -171616.8 (not concave)
Iteration 4: Log pseudolikelihood = -154870.23 (not concave)
Iteration 5: Log pseudolikelihood = -150860.02 (not concave)
Iteration 6: Log pseudolikelihood = -148049.44 (not concave)
Iteration 7: Log pseudolikelihood = -146605.44 (not concave)
Iteration 8: Log pseudolikelihood = -146196.14 (not concave)
Iteration 9: Log pseudolikelihood = -145765.13 (not concave)
Iteration 10: Log pseudolikelihood = -145556.69 (not concave)
Iteration 11: Log pseudolikelihood = -145206.07 (not concave)
Iteration 12: Log pseudolikelihood = -144949.78 (not concave)
Iteration 13: Log pseudolikelihood = -144824.78 (not concave)
Iteration 14: Log pseudolikelihood = -144718.13 (not concave)
Iteration 15: Log pseudolikelihood = -144622.65 (not concave)
Iteration 16: Log pseudolikelihood = -144532.25 (not concave)
Iteration 17: Log pseudolikelihood = -144449.28 (not concave)
Iteration 18: Log pseudolikelihood = -144372.85 (not concave)
Iteration 19: Log pseudolikelihood = -144301.29 (not concave)
Iteration 20: Log pseudolikelihood = -144130.89 (not concave)
Iteration 21: Log pseudolikelihood = -143982.46 (not concave)
Iteration 22: Log pseudolikelihood = -143850.67 (not concave)
Iteration 23: Log pseudolikelihood = -143749.92 (not concave)
Iteration 24: Log pseudolikelihood = -143668.96 (not concave)
Iteration 25: Log pseudolikelihood = -143597.47 (not concave)
Iteration 26: Log pseudolikelihood = -143526.22 (not concave)
Iteration 27: Log pseudolikelihood = -143399.94 (not concave)
Iteration 28: Log pseudolikelihood = -143275.04 (not concave)
Iteration 29: Log pseudolikelihood = -143242.43 (not concave)
Iteration 30: Log pseudolikelihood = -143006.5 (not concave)
Iteration 31: Log pseudolikelihood = -142946.42 (not concave)
Iteration 32: Log pseudolikelihood = -142925.58 (not concave)
Iteration 33: Log pseudolikelihood = -142860.21 (not concave)
Iteration 34: Log pseudolikelihood = -142785.92 (not concave)
Iteration 35: Log pseudolikelihood = -142737.35 (not concave)
Iteration 36: Log pseudolikelihood = -142724.73 (not concave)
Iteration 37: Log pseudolikelihood = -142681.42 (not concave)
Iteration 38: Log pseudolikelihood = -142657.63 (not concave)
Iteration 39: Log pseudolikelihood = -142643.87 (not concave)
Iteration 40: Log pseudolikelihood = -142640.37 (not concave)
Iteration 41: Log pseudolikelihood = -142640.06 (not concave)
Iteration 42: Log pseudolikelihood = -142640.04 (not concave)
Iteration 43: Log pseudolikelihood = -142640.04 (not concave)

```

Stereotype logistic regression

Number of obs = 1,708

Log pseudolikelihood = -142640.04

Wald chi2(21) = 211.81

Prob > chi2 = 0.0000

( 1) [phi1\_1]\_cons = 1

|                |          | sphus         | Coefficient | Robust<br>std. err. | z     | P> z  | [95% con  |
|----------------|----------|---------------|-------------|---------------------|-------|-------|-----------|
| > f. interval] |          |               |             |                     |       |       |           |
| >              | 1.072515 | age           | .4206958    | .3325669            | 1.26  | 0.206 | -.2311233 |
| >              | .0019386 | c.age#c.age   | -.0023321   | .002179             | -1.07 | 0.285 | -.0066028 |
| >              | .2029765 | yedu          | .0504685    | .0778117            | 0.65  | 0.517 | -.1020395 |
| >              | .0055731 | c.yedu#c.yedu | -.0022199   | .0039761            | -0.56 | 0.577 | -.0100129 |
| >              | .9207751 | thinc         | .2901763    | .32174              | 0.90  | 0.367 | -.3404225 |

|   |            |                                      |            |                           |       |       |            |
|---|------------|--------------------------------------|------------|---------------------------|-------|-------|------------|
| > | - .0378015 | c.thinc#c.thinc                      | - .255243  | .1109416                  | -2.30 | 0.021 | - .4726845 |
| > | .3568365   | gender<br>Female                     | - .1112494 | .2388237                  | -0.47 | 0.641 | - .5793353 |
| > | 1.041219   | household<br>Couple, both responding | .4248186   | .3144957                  | 1.35  | 0.177 | - .1915817 |
| > | 1.013627   | Multiple, at least 1 responding      | .1988463   | .4157121                  | 0.48  | 0.632 | - .6159345 |
| > | 4.770658   | cjs<br>Not applicable                | 2.301778   | 1.259656                  | 1.83  | 0.068 | - .1671012 |
| > | 1.212165   | Retired                              | .3328411   | .4486431                  | 0.74  | 0.458 | - .5464832 |
| > | 4.505932   | Unemployed                           | 3.442195   | .5427329                  | 6.34  | 0.000 | 2.378458   |
| > | 7.166549   | Permanently sick or disabled         | 4.07301    | 1.578365                  | 2.58  | 0.010 | .97947     |
| > | 1.416403   | Homemaker                            | .2674438   | .5862143                  | 0.46  | 0.648 | - .8815151 |
| > | 11.33436   | Other                                | 2.761591   | 4.37394                   | 0.63  | 0.528 | -5.811175  |
| > | 4.452524   | gali<br>Limited                      | 3.741517   | .3627656                  | 10.31 | 0.000 | 3.030509   |
| > | .3843053   | lifesat                              | -1.984826  | 1.208763                  | -1.64 | 0.101 | -4.353957  |
| > | .2062265   | c.lifesat#c.lifesat                  | .060735    | .0742317                  | 0.82  | 0.413 | - .0847565 |
| > | .6619023   | naly<br>At least 1                   | - .427685  | .5559221                  | -0.77 | 0.442 | -1.517272  |
|   |            | Not applicable                       | 0          | (omitted)                 |       |       |            |
| > | 1.50272    | rhfo<br>At least 1                   | .7936163   | .3617941                  | 2.19  | 0.028 | .0845129   |
| > | - .4315955 | ghfo<br>At least 1                   | - .9753656 | .2774388                  | -3.52 | 0.000 | -1.519136  |
| > | .876135    | rhfo#ghfo<br>At least 1#At least 1   | - .2690095 | .5842681                  | -0.46 | 0.645 | -1.414154  |
|   |            | country<br>Switzerland               | 0          | (omitted)                 |       |       |            |
| > | .9441525   | /phi1_1<br>/phi1_2                   | .8315135   | 1 (constrained)<br>.05747 | 14.47 | 0.000 | .7188744   |
| > | .6228943   | /phi1_3                              | .5351849   | .0447505                  | 11.96 | 0.000 | .4474755   |
| > | .2350355   | /phi1_4<br>/phi1_5                   | .1427583   | .0470811                  | 3.03  | 0.002 | .0504811   |
|   |            |                                      | 0          | (base outcome)            |       |       |            |
| > | 35.57757   | /theta1                              | 8.171969   | 13.98271                  | 0.58  | 0.559 | -19.23363  |
| > | 31.35427   | /theta2                              | 8.48529    | 11.66806                  | 0.73  | 0.467 | -14.38369  |
|   |            | /theta3                              | 7.246004   | 7.573361                  | 0.96  | 0.339 | -7.597511  |

```

>      22.08952
>      7.251453
      /theta4 |      3.055195      2.140987      1.43      0.154      -1.141062
      /theta5 |              0 (base outcome)

```

---

(sphus=Poor is the base outcome)  
results\_impl\_sloCountry\_sup\_1.xls  
 dir : seeout

note: **2.naly** omitted because of collinearity.  
 note: **23.country** omitted because of collinearity.

```

Iteration 0: Log pseudolikelihood = -1592729.3 (not concave)
Iteration 1: Log pseudolikelihood = -1209703.2 (not concave)
Iteration 2: Log pseudolikelihood = -539801.87 (not concave)
Iteration 3: Log pseudolikelihood = -444386.38 (not concave)
Iteration 4: Log pseudolikelihood = -330919.4 (not concave)
Iteration 5: Log pseudolikelihood = -258018.91 (not concave)
Iteration 6: Log pseudolikelihood = -231676.55 (not concave)
Iteration 7: Log pseudolikelihood = -206225.19 (not concave)
Iteration 8: Log pseudolikelihood = -198759.64 (not concave)
Iteration 9: Log pseudolikelihood = -195162.7 (not concave)
Iteration 10: Log pseudolikelihood = -191496.84 (not concave)
Iteration 11: Log pseudolikelihood = -190604.36 (not concave)
Iteration 12: Log pseudolikelihood = -188836.37 (not concave)
Iteration 13: Log pseudolikelihood = -188144.1 (not concave)
Iteration 14: Log pseudolikelihood = -187628.57 (not concave)
Iteration 15: Log pseudolikelihood = -187146.25 (not concave)
Iteration 16: Log pseudolikelihood = -184949.02 (not concave)
Iteration 17: Log pseudolikelihood = -183191.38 (not concave)
Iteration 18: Log pseudolikelihood = -183073.67 (not concave)
Iteration 19: Log pseudolikelihood = -182938.71 (not concave)
Iteration 20: Log pseudolikelihood = -182851 (not concave)
Iteration 21: Log pseudolikelihood = -182772.77 (not concave)
Iteration 22: Log pseudolikelihood = -182691.47 (not concave)
Iteration 23: Log pseudolikelihood = -182618.91 (not concave)
Iteration 24: Log pseudolikelihood = -182550.47 (not concave)
Iteration 25: Log pseudolikelihood = -182480.58 (not concave)
Iteration 26: Log pseudolikelihood = -182412.68 (not concave)
Iteration 27: Log pseudolikelihood = -182346.98 (not concave)
Iteration 28: Log pseudolikelihood = -182281.9 (not concave)
Iteration 29: Log pseudolikelihood = -182217.27 (not concave)
Iteration 30: Log pseudolikelihood = -182153.57 (not concave)
Iteration 31: Log pseudolikelihood = -182090.49 (not concave)
Iteration 32: Log pseudolikelihood = -182027.89 (not concave)
Iteration 33: Log pseudolikelihood = -181965.66 (not concave)
Iteration 34: Log pseudolikelihood = -181903.88 (not concave)
Iteration 35: Log pseudolikelihood = -181842.4 (not concave)
Iteration 36: Log pseudolikelihood = -181781.21 (not concave)
Iteration 37: Log pseudolikelihood = -181720.23 (not concave)
Iteration 38: Log pseudolikelihood = -181659.49 (not concave)
Iteration 39: Log pseudolikelihood = -181598.9 (not concave)
Iteration 40: Log pseudolikelihood = -181538.48 (not concave)
Iteration 41: Log pseudolikelihood = -181478.16 (not concave)
Iteration 42: Log pseudolikelihood = -181417.95 (not concave)
Iteration 43: Log pseudolikelihood = -181357.81 (not concave)
Iteration 44: Log pseudolikelihood = -181297.75 (not concave)
Iteration 45: Log pseudolikelihood = -181237.72 (not concave)
Iteration 46: Log pseudolikelihood = -181177.73 (not concave)
Iteration 47: Log pseudolikelihood = -181117.76 (not concave)
Iteration 48: Log pseudolikelihood = -181057.8 (not concave)
Iteration 49: Log pseudolikelihood = -180997.84 (not concave)
Iteration 50: Log pseudolikelihood = -180937.86 (not concave)
Iteration 51: Log pseudolikelihood = -180877.88 (not concave)
Iteration 52: Log pseudolikelihood = -180817.86 (not concave)
Iteration 53: Log pseudolikelihood = -180757.82 (not concave)
Iteration 54: Log pseudolikelihood = -180697.74 (not concave)
Iteration 55: Log pseudolikelihood = -180637.63 (not concave)
Iteration 56: Log pseudolikelihood = -180577.48 (not concave)
Iteration 57: Log pseudolikelihood = -180517.28 (not concave)
Iteration 58: Log pseudolikelihood = -180457.04 (not concave)
Iteration 59: Log pseudolikelihood = -180396.76 (not concave)

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Iteration 60: Log pseudolikelihood = -180336.42 (not concave)
Iteration 61: Log pseudolikelihood = -180276.06 (not concave)
Iteration 62: Log pseudolikelihood = -180215.64 (not concave)
Iteration 63: Log pseudolikelihood = -180155.19 (not concave)
Iteration 64: Log pseudolikelihood = -180094.7 (not concave)
Iteration 65: Log pseudolikelihood = -180034.18 (not concave)
Iteration 66: Log pseudolikelihood = -179973.63 (not concave)
Iteration 67: Log pseudolikelihood = -179913.07 (not concave)
Iteration 68: Log pseudolikelihood = -179852.47 (not concave)
Iteration 69: Log pseudolikelihood = -179791.88 (not concave)
Iteration 70: Log pseudolikelihood = -179731.27 (not concave)
Iteration 71: Log pseudolikelihood = -179670.69 (not concave)
Iteration 72: Log pseudolikelihood = -179610.1 (not concave)
Iteration 73: Log pseudolikelihood = -179549.55 (not concave)
Iteration 74: Log pseudolikelihood = -179489.02 (not concave)
Iteration 75: Log pseudolikelihood = -179428.55 (not concave)
Iteration 76: Log pseudolikelihood = -179368.11 (not concave)
Iteration 77: Log pseudolikelihood = -179307.77 (not concave)
Iteration 78: Log pseudolikelihood = -179247.48 (not concave)
Iteration 79: Log pseudolikelihood = -179187.31 (not concave)
Iteration 80: Log pseudolikelihood = -179127.21 (not concave)
Iteration 81: Log pseudolikelihood = -179067.27 (not concave)
Iteration 82: Log pseudolikelihood = -179007.43 (not concave)
Iteration 83: Log pseudolikelihood = -178947.77 (not concave)
Iteration 84: Log pseudolikelihood = -178888.25 (not concave)
Iteration 85: Log pseudolikelihood = -178828.94 (not concave)
Iteration 86: Log pseudolikelihood = -178769.8 (not concave)
Iteration 87: Log pseudolikelihood = -178710.91 (not concave)
Iteration 88: Log pseudolikelihood = -178652.22 (not concave)
Iteration 89: Log pseudolikelihood = -178593.81 (not concave)
Iteration 90: Log pseudolikelihood = -178535.64 (not concave)
Iteration 91: Log pseudolikelihood = -178477.8 (not concave)
Iteration 92: Log pseudolikelihood = -178420.23 (not concave)
Iteration 93: Log pseudolikelihood = -178363.02 (not concave)
Iteration 94: Log pseudolikelihood = -178306.13 (not concave)
Iteration 95: Log pseudolikelihood = -178249.64 (not concave)
Iteration 96: Log pseudolikelihood = -178193.5 (not concave)
Iteration 97: Log pseudolikelihood = -178137.8 (not concave)
Iteration 98: Log pseudolikelihood = -178082.5 (not concave)
Iteration 99: Log pseudolikelihood = -178027.67 (not concave)
Iteration 100: Log pseudolikelihood = -177973.28 (not concave)
Iteration 101: Log pseudolikelihood = -177919.4 (not concave)
Iteration 102: Log pseudolikelihood = -177866 (not concave)
Iteration 103: Log pseudolikelihood = -177813.16 (not concave)
Iteration 104: Log pseudolikelihood = -177760.83 (not concave)
Iteration 105: Log pseudolikelihood = -177709.08 (not concave)
Iteration 106: Log pseudolikelihood = -177657.89 (not concave)
Iteration 107: Log pseudolikelihood = -177607.33 (not concave)
Iteration 108: Log pseudolikelihood = -177557.35 (not concave)
Iteration 109: Log pseudolikelihood = -177508.03 (not concave)
Iteration 110: Log pseudolikelihood = -177459.33 (not concave)
Iteration 111: Log pseudolikelihood = -177411.32 (not concave)
Iteration 112: Log pseudolikelihood = -177363.97 (not concave)
Iteration 113: Log pseudolikelihood = -177317.32 (not concave)
Iteration 114: Log pseudolikelihood = -177271.36 (not concave)
Iteration 115: Log pseudolikelihood = -177226.14 (not concave)
Iteration 116: Log pseudolikelihood = -176446.75 (not concave)
Iteration 117: Log pseudolikelihood = -176362.39 (not concave)
Iteration 118: Log pseudolikelihood = -176016.69 (not concave)
Iteration 119: Log pseudolikelihood = -175835.49 (not concave)
Iteration 120: Log pseudolikelihood = -175804.76 (not concave)
Iteration 121: Log pseudolikelihood = -175758.91 (not concave)
Iteration 122: Log pseudolikelihood = -175725.19 (not concave)
Iteration 123: Log pseudolikelihood = -175549.48 (not concave)
Iteration 124: Log pseudolikelihood = -175531.51 (not concave)
Iteration 125: Log pseudolikelihood = -175531.13 (not concave)
Iteration 126: Log pseudolikelihood = -175531.12 (not concave)

```

Stereotype logistic regression

Log pseudolikelihood = -175531.12

Number of obs = 2,200

Wald chi2 (22) = 246.47

Prob > chi2 = 0.0000

( 1) [phi1\_1]\_cons = 1

|                |           | sphus                                | Coefficient | Robust<br>std. err. | z     | P> z  | [95% con  |
|----------------|-----------|--------------------------------------|-------------|---------------------|-------|-------|-----------|
| > f. interval] |           |                                      |             |                     |       |       |           |
| >              | .4772798  | age                                  | -.1368731   | .3133491            | -0.44 | 0.662 | -.7510261 |
| >              | .0052759  | c.age#c.age                          | .0012515    | .0020533            | 0.61  | 0.542 | -.0027729 |
| >              | .2043647  | yedu                                 | -.0805367   | .1453605            | -0.55 | 0.580 | -.3654382 |
| >              | .0073477  | c.yedu#c.yedu                        | -.0046738   | .0061335            | -0.76 | 0.446 | -.0166952 |
| >              | -.4494634 | thinc                                | -1.023172   | .2927139            | -3.50 | 0.000 | -1.596881 |
| >              | .6216854  | c.thinc#c.thinc                      | .3911328    | .117631             | 3.33  | 0.001 | .1605802  |
| >              | .5533118  | gender<br>Female                     | .0790875    | .2419556            | 0.33  | 0.744 | -.3951368 |
| >              | 1.517209  | household<br>Couple, both responding | .9420005    | .2934793            | 3.21  | 0.001 | .3667917  |
| >              | 1.775856  | Multiple, at least 1 responding      | .8899944    | .4519787            | 1.97  | 0.049 | .0041323  |
| >              | 2.844826  | cjs<br>Not applicable                | .5233375    | 1.184454            | 0.44  | 0.659 | -1.798151 |
| >              | 2.806883  | Retired                              | 1.00491     | .9193911            | 1.09  | 0.274 | -.7970639 |
| >              | 4.549813  | Unemployed                           | .9557158    | 1.833757            | 0.52  | 0.602 | -2.638382 |
| >              | 6.502585  | Permanently sick or disabled         | 3.283247    | 1.64255             | 2.00  | 0.046 | .063909   |
| >              | 2.601123  | Homemaker                            | .6832601    | .9785195            | 0.70  | 0.485 | -1.234603 |
| >              | 4.09886   | Other                                | 1.515692    | 1.317967            | 1.15  | 0.250 | -1.067476 |
| >              | 6.476653  | gali<br>Limited                      | 5.487366    | .5047475            | 10.87 | 0.000 | 4.498079  |
| >              | .30746    | lifesat                              | -.8072887   | .5687598            | -1.42 | 0.156 | -1.922037 |
| >              | .0629842  | c.lifesat#c.lifesat                  | -.0152652   | .0399239            | -0.38 | 0.702 | -.0935146 |
| >              | .12676    | naly<br>At least 1                   | -.6685609   | .4057835            | -1.65 | 0.099 | -1.463882 |
|                |           | Not applicable                       | 0           | (omitted)           |       |       |           |
| >              | 1.459916  | rhfo<br>At least 1                   | .8896731    | .2909457            | 3.06  | 0.002 | .31943    |
| >              | -.0714127 | ghfo<br>At least 1                   | -.6811171   | .3110794            | -2.19 | 0.029 | -1.290821 |

|                       |          |          |           |                |       |           |
|-----------------------|----------|----------|-----------|----------------|-------|-----------|
| rhfo#ghto             |          |          |           |                |       |           |
| At least 1#At least 1 |          | .3059711 | .5470601  | 0.56           | 0.576 | -.7662471 |
| >                     | 1.378189 |          |           |                |       |           |
| country               |          |          |           |                |       |           |
| Belgium               |          | 0        | (omitted) |                |       |           |
| <hr/>                 |          |          |           |                |       |           |
|                       |          | /phi1_1  | 1         | (constrained)  |       |           |
|                       |          | /phi1_2  | .9157954  | .048309        | 18.96 | 0.000     |
| >                     | 1.010479 |          |           |                |       | .8211115  |
|                       |          | /phi1_3  | .67779    | .0401607       | 16.88 | 0.000     |
| >                     | .7565035 |          |           |                |       | .5990765  |
|                       |          | /phi1_4  | .3272306  | .0445201       | 7.35  | 0.000     |
| >                     | .4144883 |          |           |                |       | .2399728  |
|                       |          | /phi1_5  | 0         | (base outcome) |       |           |
| <hr/>                 |          |          |           |                |       |           |
|                       |          | /theta1  | -7.194497 | 11.79676       | -0.61 | 0.542     |
| >                     | 15.92673 |          |           |                |       | -30.31572 |
|                       |          | /theta2  | -4.963403 | 10.80044       | -0.46 | 0.646     |
| >                     | 16.20508 |          |           |                |       | -26.13189 |
|                       |          | /theta3  | -1.580205 | 7.985238       | -0.20 | 0.843     |
| >                     | 14.07057 |          |           |                |       | -17.23098 |
|                       |          | /theta4  | .224965   | 3.85992        | 0.06  | 0.954     |
| >                     | 7.790269 |          |           |                |       | -7.340339 |
|                       |          | /theta5  | 0         | (base outcome) |       |           |
| <hr/>                 |          |          |           |                |       |           |

(sphus=Poor is the base outcome)

results\_impl\_sloCountry\_sup\_1.xls

dir : seeout

note: 2.naly omitted because of collinearity.

note: 28.country omitted because of collinearity.

```

Iteration 0: Log pseudolikelihood = -906160.95 (not concave)
Iteration 1: Log pseudolikelihood = -747421.89 (not concave)
Iteration 2: Log pseudolikelihood = -491436.82 (not concave)
Iteration 3: Log pseudolikelihood = -389617.8 (not concave)
Iteration 4: Log pseudolikelihood = -306081.8 (not concave)
Iteration 5: Log pseudolikelihood = -276412.78 (not concave)
Iteration 6: Log pseudolikelihood = -270878.95 (not concave)
Iteration 7: Log pseudolikelihood = -260747.03 (not concave)
Iteration 8: Log pseudolikelihood = -258578.61 (not concave)
Iteration 9: Log pseudolikelihood = -256209.93 (not concave)
Iteration 10: Log pseudolikelihood = -253607.38 (not concave)
Iteration 11: Log pseudolikelihood = -252713.05 (not concave)
Iteration 12: Log pseudolikelihood = -252364.57 (not concave)
Iteration 13: Log pseudolikelihood = -251916.62 (not concave)
Iteration 14: Log pseudolikelihood = -251476.41 (not concave)
Iteration 15: Log pseudolikelihood = -247205.02 (not concave)
Iteration 16: Log pseudolikelihood = -246342.26 (not concave)
Iteration 17: Log pseudolikelihood = -246012.67 (not concave)
Iteration 18: Log pseudolikelihood = -245681.16 (not concave)
Iteration 19: Log pseudolikelihood = -245313.6 (not concave)
Iteration 20: Log pseudolikelihood = -244796.36 (not concave)
Iteration 21: Log pseudolikelihood = -244275.54 (not concave)
Iteration 22: Log pseudolikelihood = -243963.38 (not concave)
Iteration 23: Log pseudolikelihood = -243591.05 (not concave)
Iteration 24: Log pseudolikelihood = -243350.31 (not concave)
Iteration 25: Log pseudolikelihood = -243128.64 (not concave)
Iteration 26: Log pseudolikelihood = -242867.6 (not concave)
Iteration 27: Log pseudolikelihood = -242492.86 (not concave)
Iteration 28: Log pseudolikelihood = -242331.42 (not concave)
Iteration 29: Log pseudolikelihood = -242189.88 (not concave)
Iteration 30: Log pseudolikelihood = -242045.8 (not concave)
Iteration 31: Log pseudolikelihood = -241919.57 (not concave)
Iteration 32: Log pseudolikelihood = -241801.76 (not concave)
Iteration 33: Log pseudolikelihood = -241685.64 (not concave)
Iteration 34: Log pseudolikelihood = -241579.08 (not concave)
Iteration 35: Log pseudolikelihood = -241481.17 (not concave)
Iteration 36: Log pseudolikelihood = -241389.1 (not concave)

```

Iteration 37: Log pseudolikelihood = -241303.59 (not concave)  
Iteration 38: Log pseudolikelihood = -241225.03 (not concave)  
Iteration 39: Log pseudolikelihood = -241152.69 (not concave)  
Iteration 40: Log pseudolikelihood = -241086.3 (not concave)  
Iteration 41: Log pseudolikelihood = -241025.44 (not concave)  
Iteration 42: Log pseudolikelihood = -240969.74 (not concave)  
Iteration 43: Log pseudolikelihood = -240918.7 (not concave)  
Iteration 44: Log pseudolikelihood = -240871.96 (not concave)  
Iteration 45: Log pseudolikelihood = -240829.03 (not concave)  
Iteration 46: Log pseudolikelihood = -240789.58 (not concave)  
Iteration 47: Log pseudolikelihood = -240753.18 (not concave)  
Iteration 48: Log pseudolikelihood = -240719.56 (not concave)  
Iteration 49: Log pseudolikelihood = -240688.34 (not concave)  
Iteration 50: Log pseudolikelihood = -240659.31 (not concave)  
Iteration 51: Log pseudolikelihood = -240632.19 (not concave)  
Iteration 52: Log pseudolikelihood = -240606.8 (not concave)  
Iteration 53: Log pseudolikelihood = -240582.92 (not concave)  
Iteration 54: Log pseudolikelihood = -240560.41 (not concave)  
Iteration 55: Log pseudolikelihood = -240539.12 (not concave)  
Iteration 56: Log pseudolikelihood = -240518.93 (not concave)  
Iteration 57: Log pseudolikelihood = -240499.72 (not concave)  
Iteration 58: Log pseudolikelihood = -240481.42 (not concave)  
Iteration 59: Log pseudolikelihood = -240463.92 (not concave)  
Iteration 60: Log pseudolikelihood = -240447.17 (not concave)  
Iteration 61: Log pseudolikelihood = -240431.1 (not concave)  
Iteration 62: Log pseudolikelihood = -240415.65 (not concave)  
Iteration 63: Log pseudolikelihood = -240400.76 (not concave)  
Iteration 64: Log pseudolikelihood = -240386.41 (not concave)  
Iteration 65: Log pseudolikelihood = -240372.54 (not concave)  
Iteration 66: Log pseudolikelihood = -240359.12 (not concave)  
Iteration 67: Log pseudolikelihood = -240346.12 (not concave)  
Iteration 68: Log pseudolikelihood = -240333.52 (not concave)  
Iteration 69: Log pseudolikelihood = -240321.27 (not concave)  
Iteration 70: Log pseudolikelihood = -240309.38 (not concave)  
Iteration 71: Log pseudolikelihood = -240297.8 (not concave)  
Iteration 72: Log pseudolikelihood = -240286.52 (not concave)  
Iteration 73: Log pseudolikelihood = -240275.53 (not concave)  
Iteration 74: Log pseudolikelihood = -240264.82 (not concave)  
Iteration 75: Log pseudolikelihood = -240254.35 (not concave)  
Iteration 76: Log pseudolikelihood = -240244.13 (not concave)  
Iteration 77: Log pseudolikelihood = -240234.14 (not concave)  
Iteration 78: Log pseudolikelihood = -240224.37 (not concave)  
Iteration 79: Log pseudolikelihood = -240214.8 (not concave)  
Iteration 80: Log pseudolikelihood = -240205.44 (not concave)  
Iteration 81: Log pseudolikelihood = -240196.27 (not concave)  
Iteration 82: Log pseudolikelihood = -240187.29 (not concave)  
Iteration 83: Log pseudolikelihood = -240178.48 (not concave)  
Iteration 84: Log pseudolikelihood = -240169.85 (not concave)  
Iteration 85: Log pseudolikelihood = -240161.38 (not concave)  
Iteration 86: Log pseudolikelihood = -240153.07 (not concave)  
Iteration 87: Log pseudolikelihood = -240144.91 (not concave)  
Iteration 88: Log pseudolikelihood = -240136.91 (not concave)  
Iteration 89: Log pseudolikelihood = -240129.04 (not concave)  
Iteration 90: Log pseudolikelihood = -240121.32 (not concave)  
Iteration 91: Log pseudolikelihood = -240113.74 (not concave)  
Iteration 92: Log pseudolikelihood = -240106.29 (not concave)  
Iteration 93: Log pseudolikelihood = -240098.96 (not concave)  
Iteration 94: Log pseudolikelihood = -240091.77 (not concave)  
Iteration 95: Log pseudolikelihood = -240084.69 (not concave)  
Iteration 96: Log pseudolikelihood = -240077.74 (not concave)  
Iteration 97: Log pseudolikelihood = -240070.9 (not concave)  
Iteration 98: Log pseudolikelihood = -240064.18 (not concave)  
Iteration 99: Log pseudolikelihood = -240057.57 (not concave)  
Iteration 100: Log pseudolikelihood = -240051.07 (not concave)  
Iteration 101: Log pseudolikelihood = -240044.68 (not concave)  
Iteration 102: Log pseudolikelihood = -240038.4 (not concave)  
Iteration 103: Log pseudolikelihood = -240032.21 (not concave)  
Iteration 104: Log pseudolikelihood = -240026.13 (not concave)  
Iteration 105: Log pseudolikelihood = -240020.15 (not concave)  
Iteration 106: Log pseudolikelihood = -240014.26 (not concave)  
Iteration 107: Log pseudolikelihood = -240008.47 (not concave)  
Iteration 108: Log pseudolikelihood = -240002.78 (not concave)

Iteration 109: Log pseudolikelihood = -239997.17 (not concave)  
Iteration 110: Log pseudolikelihood = -239991.66 (not concave)  
Iteration 111: Log pseudolikelihood = -239986.23 (not concave)  
Iteration 112: Log pseudolikelihood = -239980.9 (not concave)  
Iteration 113: Log pseudolikelihood = -239975.64 (not concave)  
Iteration 114: Log pseudolikelihood = -239970.48 (not concave)  
Iteration 115: Log pseudolikelihood = -239965.39 (not concave)  
Iteration 116: Log pseudolikelihood = -239960.39 (not concave)  
Iteration 117: Log pseudolikelihood = -239955.47 (not concave)  
Iteration 118: Log pseudolikelihood = -239950.62 (not concave)  
Iteration 119: Log pseudolikelihood = -239945.86 (not concave)  
Iteration 120: Log pseudolikelihood = -239941.17 (not concave)  
Iteration 121: Log pseudolikelihood = -239936.55 (not concave)  
Iteration 122: Log pseudolikelihood = -239932.01 (not concave)  
Iteration 123: Log pseudolikelihood = -239927.54 (not concave)  
Iteration 124: Log pseudolikelihood = -239923.14 (not concave)  
Iteration 125: Log pseudolikelihood = -239918.81 (not concave)  
Iteration 126: Log pseudolikelihood = -239914.56 (not concave)  
Iteration 127: Log pseudolikelihood = -239910.36 (not concave)  
Iteration 128: Log pseudolikelihood = -239906.24 (not concave)  
Iteration 129: Log pseudolikelihood = -239902.18 (not concave)  
Iteration 130: Log pseudolikelihood = -239898.19 (not concave)  
Iteration 131: Log pseudolikelihood = -239894.26 (not concave)  
Iteration 132: Log pseudolikelihood = -239890.39 (not concave)  
Iteration 133: Log pseudolikelihood = -239886.58 (not concave)  
Iteration 134: Log pseudolikelihood = -239882.84 (not concave)  
Iteration 135: Log pseudolikelihood = -239879.15 (not concave)  
Iteration 136: Log pseudolikelihood = -239875.53 (not concave)  
Iteration 137: Log pseudolikelihood = -239871.96 (not concave)  
Iteration 138: Log pseudolikelihood = -239868.44 (not concave)  
Iteration 139: Log pseudolikelihood = -239864.99 (not concave)  
Iteration 140: Log pseudolikelihood = -239861.58 (not concave)  
Iteration 141: Log pseudolikelihood = -239858.23 (not concave)  
Iteration 142: Log pseudolikelihood = -239854.94 (not concave)  
Iteration 143: Log pseudolikelihood = -239851.7 (not concave)  
Iteration 144: Log pseudolikelihood = -239848.5 (not concave)  
Iteration 145: Log pseudolikelihood = -239845.36 (not concave)  
Iteration 146: Log pseudolikelihood = -239842.27 (not concave)  
Iteration 147: Log pseudolikelihood = -239839.23 (not concave)  
Iteration 148: Log pseudolikelihood = -239836.24 (not concave)  
Iteration 149: Log pseudolikelihood = -239833.29 (not concave)  
Iteration 150: Log pseudolikelihood = -239830.39 (not concave)  
Iteration 151: Log pseudolikelihood = -239827.53 (not concave)  
Iteration 152: Log pseudolikelihood = -239824.72 (not concave)  
Iteration 153: Log pseudolikelihood = -239821.96 (not concave)  
Iteration 154: Log pseudolikelihood = -239819.24 (not concave)  
Iteration 155: Log pseudolikelihood = -239816.56 (not concave)  
Iteration 156: Log pseudolikelihood = -239813.92 (not concave)  
Iteration 157: Log pseudolikelihood = -239811.33 (not concave)  
Iteration 158: Log pseudolikelihood = -239808.77 (not concave)  
Iteration 159: Log pseudolikelihood = -239806.26 (not concave)  
Iteration 160: Log pseudolikelihood = -239803.79 (not concave)  
Iteration 161: Log pseudolikelihood = -239801.35 (not concave)  
Iteration 162: Log pseudolikelihood = -239798.95 (not concave)  
Iteration 163: Log pseudolikelihood = -239796.59 (not concave)  
Iteration 164: Log pseudolikelihood = -239794.27 (not concave)  
Iteration 165: Log pseudolikelihood = -239791.98 (not concave)  
Iteration 166: Log pseudolikelihood = -239789.73 (not concave)  
Iteration 167: Log pseudolikelihood = -239787.52 (not concave)  
Iteration 168: Log pseudolikelihood = -239785.34 (not concave)  
Iteration 169: Log pseudolikelihood = -239783.19 (not concave)  
Iteration 170: Log pseudolikelihood = -239781.08 (not concave)  
Iteration 171: Log pseudolikelihood = -239779 (not concave)  
Iteration 172: Log pseudolikelihood = -239776.95 (not concave)  
Iteration 173: Log pseudolikelihood = -239774.93 (not concave)  
Iteration 174: Log pseudolikelihood = -239772.95 (not concave)  
Iteration 175: Log pseudolikelihood = -239770.99 (not concave)  
Iteration 176: Log pseudolikelihood = -239769.07 (not concave)  
Iteration 177: Log pseudolikelihood = -239767.17 (not concave)  
Iteration 178: Log pseudolikelihood = -239765.31 (not concave)  
Iteration 179: Log pseudolikelihood = -239763.47 (not concave)  
Iteration 180: Log pseudolikelihood = -239761.67 (not concave)

Iteration 181: Log pseudolikelihood = **-239759.89** (not concave)  
Iteration 182: Log pseudolikelihood = **-239758.13** (not concave)  
Iteration 183: Log pseudolikelihood = **-239756.41** (not concave)  
Iteration 184: Log pseudolikelihood = **-239754.71** (not concave)  
Iteration 185: Log pseudolikelihood = **-239753.04** (not concave)  
Iteration 186: Log pseudolikelihood = **-239751.39** (not concave)  
Iteration 187: Log pseudolikelihood = **-239749.77** (not concave)  
Iteration 188: Log pseudolikelihood = **-239748.17** (not concave)  
Iteration 189: Log pseudolikelihood = **-239746.59** (not concave)  
Iteration 190: Log pseudolikelihood = **-239745.05** (not concave)  
Iteration 191: Log pseudolikelihood = **-239743.52** (not concave)  
Iteration 192: Log pseudolikelihood = **-239742.02** (not concave)  
Iteration 193: Log pseudolikelihood = **-239740.54** (not concave)  
Iteration 194: Log pseudolikelihood = **-239739.08** (not concave)  
Iteration 195: Log pseudolikelihood = **-239737.65** (not concave)  
Iteration 196: Log pseudolikelihood = **-239736.23** (not concave)  
Iteration 197: Log pseudolikelihood = **-239734.84** (not concave)  
Iteration 198: Log pseudolikelihood = **-239733.47** (not concave)  
Iteration 199: Log pseudolikelihood = **-239732.12** (not concave)  
Iteration 200: Log pseudolikelihood = **-239730.79** (not concave)  
Iteration 201: Log pseudolikelihood = **-239729.48** (not concave)  
Iteration 202: Log pseudolikelihood = **-239728.19** (not concave)  
Iteration 203: Log pseudolikelihood = **-239726.92** (not concave)  
Iteration 204: Log pseudolikelihood = **-239725.67** (not concave)  
Iteration 205: Log pseudolikelihood = **-239724.43** (not concave)  
Iteration 206: Log pseudolikelihood = **-239723.22** (not concave)  
Iteration 207: Log pseudolikelihood = **-239722.02** (not concave)  
Iteration 208: Log pseudolikelihood = **-239720.84** (not concave)  
Iteration 209: Log pseudolikelihood = **-239719.68** (not concave)  
Iteration 210: Log pseudolikelihood = **-239718.53** (not concave)  
Iteration 211: Log pseudolikelihood = **-239717.41** (not concave)  
Iteration 212: Log pseudolikelihood = **-239716.29** (not concave)  
Iteration 213: Log pseudolikelihood = **-239715.2** (not concave)  
Iteration 214: Log pseudolikelihood = **-239714.12** (not concave)  
Iteration 215: Log pseudolikelihood = **-239713.06** (not concave)  
Iteration 216: Log pseudolikelihood = **-239712.01** (not concave)  
Iteration 217: Log pseudolikelihood = **-239710.98** (not concave)  
Iteration 218: Log pseudolikelihood = **-239709.96** (not concave)  
Iteration 219: Log pseudolikelihood = **-239708.96** (not concave)  
Iteration 220: Log pseudolikelihood = **-239707.97** (not concave)  
Iteration 221: Log pseudolikelihood = **-239707** (not concave)  
Iteration 222: Log pseudolikelihood = **-239706.04** (not concave)  
Iteration 223: Log pseudolikelihood = **-239705.09** (not concave)  
Iteration 224: Log pseudolikelihood = **-239704.16** (not concave)  
Iteration 225: Log pseudolikelihood = **-239703.24** (not concave)  
Iteration 226: Log pseudolikelihood = **-239702.34** (not concave)  
Iteration 227: Log pseudolikelihood = **-239696.04** (not concave)  
Iteration 228: Log pseudolikelihood = **-239694.57** (not concave)  
Iteration 229: Log pseudolikelihood = **-239693.52** (not concave)  
Iteration 230: Log pseudolikelihood = **-239692.5** (not concave)  
Iteration 231: Log pseudolikelihood = **-239690.99** (not concave)  
Iteration 232: Log pseudolikelihood = **-239689.52** (not concave)  
Iteration 233: Log pseudolikelihood = **-239688.65** (not concave)  
Iteration 234: Log pseudolikelihood = **-239687.53** (not concave)  
Iteration 235: Log pseudolikelihood = **-239687.02** (not concave)  
Iteration 236: Log pseudolikelihood = **-239686.24** (not concave)  
Iteration 237: Log pseudolikelihood = **-239685.46** (not concave)  
Iteration 238: Log pseudolikelihood = **-239683.12** (not concave)  
Iteration 239: Log pseudolikelihood = **-239680.75** (not concave)  
Iteration 240: Log pseudolikelihood = **-239680.3** (not concave)  
Iteration 241: Log pseudolikelihood = **-239679.88** (not concave)  
Iteration 242: Log pseudolikelihood = **-239679.31** (not concave)  
Iteration 243: Log pseudolikelihood = **-239678.63** (not concave)  
Iteration 244: Log pseudolikelihood = **-239678.22** (not concave)  
Iteration 245: Log pseudolikelihood = **-239677.8** (not concave)  
Iteration 246: Log pseudolikelihood = **-239658.15** (not concave)  
Iteration 247: Log pseudolikelihood = **-239643.5** (not concave)  
Iteration 248: Log pseudolikelihood = **-239643.36** (not concave)  
Iteration 249: Log pseudolikelihood = **-239643.1** (not concave)  
Iteration 250: Log pseudolikelihood = **-239640.9** (not concave)  
Iteration 251: Log pseudolikelihood = **-239639.89** (not concave)  
Iteration 252: Log pseudolikelihood = **-239639.66** (not concave)

```

Iteration 253: Log pseudolikelihood = -239639.52 (not concave)
Iteration 254: Log pseudolikelihood = -239639.38 (not concave)
Iteration 255: Log pseudolikelihood = -239638.95 (not concave)
Iteration 256: Log pseudolikelihood = -239638.53 (not concave)
Iteration 257: Log pseudolikelihood = -239638.31 (not concave)
Iteration 258: Log pseudolikelihood = -239638.17 (not concave)
Iteration 259: Log pseudolikelihood = -239638.05 (not concave)
Iteration 260: Log pseudolikelihood = -239637.92 (not concave)
Iteration 261: Log pseudolikelihood = -239637.8 (not concave)
Iteration 262: Log pseudolikelihood = -239637.67 (not concave)
Iteration 263: Log pseudolikelihood = -239637.55 (not concave)
Iteration 264: Log pseudolikelihood = -239637.43 (not concave)
Iteration 265: Log pseudolikelihood = -239637.31 (not concave)
Iteration 266: Log pseudolikelihood = -239637.19 (not concave)
Iteration 267: Log pseudolikelihood = -239637.07 (not concave)
Iteration 268: Log pseudolikelihood = -239636.95 (not concave)
Iteration 269: Log pseudolikelihood = -239636.83 (not concave)
Iteration 270: Log pseudolikelihood = -239636.71 (not concave)
Iteration 271: Log pseudolikelihood = -239636.6 (not concave)
Iteration 272: Log pseudolikelihood = -239636.48 (not concave)
Iteration 273: Log pseudolikelihood = -239636.36 (not concave)
Iteration 274: Log pseudolikelihood = -239636.25 (not concave)
Iteration 275: Log pseudolikelihood = -239636.13 (not concave)
Iteration 276: Log pseudolikelihood = -239636.02 (not concave)
Iteration 277: Log pseudolikelihood = -239635.91 (not concave)
Iteration 278: Log pseudolikelihood = -239635.79 (not concave)
Iteration 279: Log pseudolikelihood = -239635.68 (not concave)
Iteration 280: Log pseudolikelihood = -239635.57 (not concave)
Iteration 281: Log pseudolikelihood = -239635.46 (not concave)
Iteration 282: Log pseudolikelihood = -239635.35 (not concave)
Iteration 283: Log pseudolikelihood = -239635.24 (not concave)
Iteration 284: Log pseudolikelihood = -239635.13 (not concave)
Iteration 285: Log pseudolikelihood = -239635.02 (not concave)
Iteration 286: Log pseudolikelihood = -239634.91 (not concave)
Iteration 287: Log pseudolikelihood = -239634.8 (not concave)
Iteration 288: Log pseudolikelihood = -239634.7 (not concave)
Iteration 289: Log pseudolikelihood = -239634.59 (not concave)
Iteration 290: Log pseudolikelihood = -239634.48 (not concave)
Iteration 291: Log pseudolikelihood = -239634.38 (not concave)
Iteration 292: Log pseudolikelihood = -239634.27 (not concave)
Iteration 293: Log pseudolikelihood = -239634.17 (not concave)
Iteration 294: Log pseudolikelihood = -239634.07 (not concave)
Iteration 295: Log pseudolikelihood = -239633.96 (not concave)
Iteration 296: Log pseudolikelihood = -239633.86 (not concave)
Iteration 297: Log pseudolikelihood = -239633.76 (not concave)
Iteration 298: Log pseudolikelihood = -239633.65 (not concave)
Iteration 299: Log pseudolikelihood = -239633.55 (not concave)
Iteration 300: Log pseudolikelihood = -239633.45 (not concave)

```

**convergence not achieved**

Stereotype logistic regression

Number of obs = **2,562**

Log pseudolikelihood = **-3006.336**

Wald chi2(21) = **1784.74**

Prob > chi2 = **0.0000**

( 1) [phi1\_1]\_cons = 1

|                |           | sphus         | Coefficient | Robust<br>std. err. | z     | P> z  | [95% con  |
|----------------|-----------|---------------|-------------|---------------------|-------|-------|-----------|
| > f. interval] |           |               |             |                     |       |       |           |
| >              | .3314366  | age           | .2233731    | .0551355            | 4.05  | 0.000 | .1153096  |
| >              | -.0003179 | c.age#c.age   | -.0010942   | .0003961            | -2.76 | 0.006 | -.0018706 |
| >              | .2792577  | yedu          | .0108519    | .1369442            | 0.08  | 0.937 | -.2575539 |
|                |           | c.yedu#c.yedu | -.0056469   | .0056586            | -1.00 | 0.318 | -.0167375 |

|   |           |                                      |               |                           |       |       |           |
|---|-----------|--------------------------------------|---------------|---------------------------|-------|-------|-----------|
| > | .0054438  |                                      |               |                           |       |       |           |
| > | 4.732517  | thinc                                | .3070441      | 2.257936                  | 0.14  | 0.892 | -4.118429 |
| > | 5.392657  | c.thinc#c.thinc                      | 1.924354      | 1.769575                  | 1.09  | 0.277 | -1.543949 |
| > | -.0760748 | gender<br>Female                     | -.481833      | .2070233                  | -2.33 | 0.020 | -.8875912 |
| > | .8424794  | household<br>Couple, both responding | .2726411      | .2907392                  | 0.94  | 0.348 | -.2971973 |
| > | .8502107  | Multiple, at least 1 responding      | -.2951115     | .5843588                  | -0.51 | 0.614 | -1.440434 |
| > | 4.974906  | cjs<br>Not applicable                | 2.638823      | 1.191901                  | 2.21  | 0.027 | .3027409  |
| > | 3.725444  | Retired                              | 1.916147      | .9231279                  | 2.08  | 0.038 | .1068498  |
| > | .         | Unemployed                           | 343.1497      | .                         | .     | .     | .         |
| > | 8.692949  | Permanently sick or disabled         | 1.964088      | 3.433155                  | 0.57  | 0.567 | -4.764772 |
| > | -.8533684 | Homemaker                            | -3.039703     | 1.115497                  | -2.72 | 0.006 | -5.226038 |
| > | 5.556126  | Other                                | 3.593683      | 1.001265                  | 3.59  | 0.000 | 1.631241  |
| > | 5.212413  | gali<br>Limited                      | 4.498656      | .3641684                  | 12.35 | 0.000 | 3.784899  |
| > | -.3705635 | lifesat                              | -1.184682     | .4153742                  | -2.85 | 0.004 | -1.9988   |
| > | .1075908  | c.lifesat#c.lifesat                  | .0503694      | .0291951                  | 1.73  | 0.084 | -.0068519 |
| > | .0316435  | naly<br>At least 1                   | -.6917434     | .3690817                  | -1.87 | 0.061 | -1.41513  |
|   |           | Not applicable                       | 0             | (omitted)                 |       |       |           |
| > | 1.536213  | rhfo<br>At least 1                   | 1.063119      | .2413792                  | 4.40  | 0.000 | .5900243  |
| > | -.5495323 | ghfo<br>At least 1                   | -1.162763     | .3128785                  | -3.72 | 0.000 | -1.775994 |
| > | .3139187  | rhfo#ghfo<br>At least 1#At least 1   | -.5245083     | .4277767                  | -1.23 | 0.220 | -1.362935 |
|   |           | country<br>Czech Republic            | 0             | (omitted)                 |       |       |           |
| > | .9412193  | /phi1_1<br>/phi1_2                   | 1<br>.8301004 | (constrained)<br>.0566944 | 14.64 | 0.000 | .7189815  |
| > | .6800555  | /phi1_3                              | .6053904      | .0380951                  | 15.89 | 0.000 | .5307254  |
| > | .4496854  | /phi1_4<br>/phi1_5                   | .3874073      | .0317752                  | 12.19 | 0.000 | .3251291  |
|   |           |                                      | 0             | (base outcome)            |       |       |           |

```

>          /theta1 |    6.640814      .      .      .      .
>          /theta2 |    7.01688    .3543349    19.80    0.000    6.322396
>    7.711363      /theta3 |    6.752999    .2701408    25.00    0.000    6.223533
>    7.282465      /theta4 |    4.926248    .2869025    17.17    0.000    4.36393
>    5.488567      /theta5 |            0 (base outcome)

```

```

(sphus=Poor is the base outcome)
convergence not achieved
r(430);

```

```
end of do-file
```

```
r(430);
```

```
1 . tabulate country
```

| Country<br>identifier | Freq.  | Percent | Cum.   |
|-----------------------|--------|---------|--------|
| Austria               | 8,220  | 12.11   | 12.11  |
| Germany               | 2,952  | 4.35    | 16.46  |
| Sweden                | 4,171  | 6.15    | 22.61  |
| Spain                 | 6,667  | 9.83    | 32.44  |
| Italy                 | 6,309  | 9.30    | 41.74  |
| France                | 8,477  | 12.49   | 54.23  |
| Denmark               | 3,605  | 5.31    | 59.54  |
| Switzerland           | 5,996  | 8.84    | 68.38  |
| Belgium               | 7,917  | 11.67   | 80.05  |
| Czech Republic        | 9,308  | 13.72   | 93.76  |
| Slovenia              | 4,232  | 6.24    | 100.00 |
| Total                 | 67,854 | 100.00  |        |

```
2 . tabulate country, nolab
```

| Country<br>identifier | Freq.  | Percent | Cum.   |
|-----------------------|--------|---------|--------|
| 11                    | 8,220  | 12.11   | 12.11  |
| 12                    | 2,952  | 4.35    | 16.46  |
| 13                    | 4,171  | 6.15    | 22.61  |
| 15                    | 6,667  | 9.83    | 32.44  |
| 16                    | 6,309  | 9.30    | 41.74  |
| 17                    | 8,477  | 12.49   | 54.23  |
| 18                    | 3,605  | 5.31    | 59.54  |
| 20                    | 5,996  | 8.84    | 68.38  |
| 23                    | 7,917  | 11.67   | 80.05  |
| 28                    | 9,308  | 13.72   | 93.76  |
| 34                    | 4,232  | 6.24    | 100.00 |
| Total                 | 67,854 | 100.00  |        |

```
3 . do "C:\Users\hansg\AppData\Local\Temp\STD1bdc_000000.tmp"
```

```

4 .      slogit sphus c.age##c.age c.yedu##c.yedu c.thinc##c.thinc i.gender ib(#1).ho
> usehold ib(#3)o2.cjs i.gali ///
>      c.lifesat##c.lifesat i.naly i.rhfo##i.ighto ib(#3).country [pweight=weight] i
> f Qyear==2015 & country==34, vce(robust)
note: 2.naly omitted because of collinearity.
note: 34.country omitted because of collinearity.

```

```

Iteration 0: Log pseudolikelihood = -752091.62 (not concave)
Iteration 1: Log pseudolikelihood = -551294.53 (not concave)
Iteration 2: Log pseudolikelihood = -248332.43 (not concave)
Iteration 3: Log pseudolikelihood = -147801.81 (not concave)
Iteration 4: Log pseudolikelihood = -127380.13 (not concave)
Iteration 5: Log pseudolikelihood = -117808.15 (not concave)
Iteration 6: Log pseudolikelihood = -115145.89 (not concave)
Iteration 7: Log pseudolikelihood = -113162.25 (not concave)
Iteration 8: Log pseudolikelihood = -111880.14 (not concave)
Iteration 9: Log pseudolikelihood = -110678.06 (not concave)
Iteration 10: Log pseudolikelihood = -108842.1 (not concave)
Iteration 11: Log pseudolikelihood = -108163.48 (not concave)
Iteration 12: Log pseudolikelihood = -107953.49 (not concave)
Iteration 13: Log pseudolikelihood = -107707.7 (not concave)
Iteration 14: Log pseudolikelihood = -107573.48 (not concave)
Iteration 15: Log pseudolikelihood = -107448.39 (not concave)
Iteration 16: Log pseudolikelihood = -107206.21 (not concave)
Iteration 17: Log pseudolikelihood = -107064.53 (not concave)
Iteration 18: Log pseudolikelihood = -106977.44 (not concave)
Iteration 19: Log pseudolikelihood = -106889.63 (not concave)
Iteration 20: Log pseudolikelihood = -106298.54 (not concave)
Iteration 21: Log pseudolikelihood = -105865.48 (not concave)
Iteration 22: Log pseudolikelihood = -105772.45 (not concave)
Iteration 23: Log pseudolikelihood = -105691.21 (not concave)
Iteration 24: Log pseudolikelihood = -105559.35 (not concave)
Iteration 25: Log pseudolikelihood = -105437.34 (not concave)
Iteration 26: Log pseudolikelihood = -104121.09 (not concave)
Iteration 27: Log pseudolikelihood = -103873.04 (not concave)
Iteration 28: Log pseudolikelihood = -103793.7 (not concave)
Iteration 29: Log pseudolikelihood = -103709.43 (not concave)
Iteration 30: Log pseudolikelihood = -103603.17 (not concave)
Iteration 31: Log pseudolikelihood = -103461.51 (not concave)
Iteration 32: Log pseudolikelihood = -103412.22 (not concave)
Iteration 33: Log pseudolikelihood = -103363.16 (not concave)
Iteration 34: Log pseudolikelihood = -103135.11 (not concave)
Iteration 35: Log pseudolikelihood = -102783.2 (not concave)
Iteration 36: Log pseudolikelihood = -102708.29 (not concave)
Iteration 37: Log pseudolikelihood = -102647.72 (not concave)
Iteration 38: Log pseudolikelihood = -102597.57 (not concave)
Iteration 39: Log pseudolikelihood = -102553.26 (not concave)
Iteration 40: Log pseudolikelihood = -102455.38 (not concave)
Iteration 41: Log pseudolikelihood = -102294.95 (not concave)
Iteration 42: Log pseudolikelihood = -102197.12 (not concave)
Iteration 43: Log pseudolikelihood = -101923.52 (not concave)
Iteration 44: Log pseudolikelihood = -101800.06 (not concave)
Iteration 45: Log pseudolikelihood = -101785.68 (not concave)
Iteration 46: Log pseudolikelihood = -101578.77 (not concave)
Iteration 47: Log pseudolikelihood = -101541.1 (not concave)
Iteration 48: Log pseudolikelihood = -101410.56 (not concave)
Iteration 49: Log pseudolikelihood = -101377.27 (not concave)
Iteration 50: Log pseudolikelihood = -101341.39 (not concave)
Iteration 51: Log pseudolikelihood = -101325.3 (not concave)
Iteration 52: Log pseudolikelihood = -101321.8 (not concave)
Iteration 53: Log pseudolikelihood = -101318.02 (not concave)
Iteration 54: Log pseudolikelihood = -101317.42 (not concave)
Iteration 55: Log pseudolikelihood = -101317.41 (not concave)

```

Stereotype logistic regression

Log pseudolikelihood = **-101317.41**

Number of obs = **1,214**

Wald chi2(22) = **96.02**

Prob > chi2 = **0.0000**

( 1) [phi1\_1]\_cons = 1

|                |           | sphus                                | Coefficient | Robust<br>std. err. | z     | P> z  | [95% con  |
|----------------|-----------|--------------------------------------|-------------|---------------------|-------|-------|-----------|
| > f. interval] |           |                                      |             |                     |       |       |           |
| >              | 1.399626  | age                                  | .454527     | .4822021            | 0.94  | 0.346 | -.4905718 |
| >              | .0037388  | c.age#c.age                          | -.0024723   | .003169             | -0.78 | 0.435 | -.0086834 |
| >              | .5204707  | yedu                                 | .1220951    | .2032566            | 0.60  | 0.548 | -.2762806 |
| >              | .0025241  | c.yedu#c.yedu                        | -.0158792   | .0093896            | -1.69 | 0.091 | -.0342825 |
| >              | .4473457  | thinc                                | -.5034596   | .4851137            | -1.04 | 0.299 | -1.454265 |
| >              | .4869579  | c.thinc#c.thinc                      | -.0213946   | .2593683            | -0.08 | 0.934 | -.5297472 |
| >              | .2914787  | gender<br>Female                     | -.3711744   | .3380945            | -1.10 | 0.272 | -1.033827 |
| >              | 1.568044  | household<br>Couple, both responding | .7194979    | .4329398            | 1.66  | 0.097 | -.1290486 |
| >              | 1.031552  | Multiple, at least 1 responding      | -.1826505   | .6195026            | -0.29 | 0.768 | -1.396853 |
| >              | 16.33556  | cjs<br>Not applicable                | 9.061753    | 3.711195            | 2.44  | 0.015 | 1.787945  |
| >              | 11.73896  | Retired                              | 5.104553    | 3.384963            | 1.51  | 0.132 | -1.529852 |
| >              | 8.999246  | Unemployed                           | 1.260964    | 3.948175            | 0.32  | 0.749 | -6.477317 |
| >              | 14.67863  | Permanently sick or disabled         | 7.226001    | 3.802434            | 1.90  | 0.057 | -.2266331 |
| >              | 12.59623  | Homemaker                            | 5.859257    | 3.437296            | 1.70  | 0.088 | -.8777195 |
| >              | 10.88121  | Other                                | 3.91157     | 3.556003            | 1.10  | 0.271 | -3.058069 |
| >              | 6.448367  | gali<br>Limited                      | 5.094515    | .6907535            | 7.38  | 0.000 | 3.740663  |
| >              | -.7730456 | lifesat                              | -1.725384   | .4858959            | -3.55 | 0.000 | -2.677722 |
| >              | .145099   | c.lifesat#c.lifesat                  | .0775961    | .0344409            | 2.25  | 0.024 | .0100931  |
| >              | -.0296738 | naly<br>At least 1                   | -.9211414   | .4548388            | -2.03 | 0.043 | -1.812609 |
|                |           | Not applicable                       | 0           | (omitted)           |       |       |           |
| >              | 2.137748  | rhfo<br>At least 1                   | 1.239544    | .4582755            | 2.70  | 0.007 | .3413408  |
| >              | 1.185963  | ghfo<br>At least 1                   | .2622827    | .4712739            | 0.56  | 0.578 | -.661397  |

|                       |          |           |                |       |       |           |
|-----------------------|----------|-----------|----------------|-------|-------|-----------|
| rhfo#ghto             |          |           |                |       |       |           |
| At least 1#At least 1 |          | -1.769904 | .9986861       | -1.77 | 0.076 | -3.727293 |
| >                     | .1874847 |           |                |       |       |           |
| country               |          |           |                |       |       |           |
| Slovenia              |          | 0         | (omitted)      |       |       |           |
| <hr/>                 |          |           |                |       |       |           |
| /phi1_1               |          | 1         | (constrained)  |       |       |           |
| /phi1_2               |          | .7888707  | .091587        | 8.61  | 0.000 | .6093636  |
| >                     | .9683779 |           |                |       |       |           |
| /phi1_3               |          | .5372745  | .0687042       | 7.82  | 0.000 | .4026168  |
| >                     | .6719322 |           |                |       |       |           |
| /phi1_4               |          | .3386082  | .0498204       | 6.80  | 0.000 | .2409621  |
| >                     | .4362544 |           |                |       |       |           |
| /phi1_5               |          | 0         | (base outcome) |       |       |           |
| <hr/>                 |          |           |                |       |       |           |
| /theta1               |          | 16.17508  | 18.93828       | 0.85  | 0.393 | -20.94326 |
| >                     | 53.29342 |           |                |       |       |           |
| /theta2               |          | 14.22055  | 15.11764       | 0.94  | 0.347 | -15.40947 |
| >                     | 43.85057 |           |                |       |       |           |
| /theta3               |          | 11.84389  | 10.50794       | 1.13  | 0.260 | -8.75129  |
| >                     | 32.43907 |           |                |       |       |           |
| /theta4               |          | 7.777635  | 6.637196       | 1.17  | 0.241 | -5.231029 |
| >                     | 20.7863  |           |                |       |       |           |
| /theta5               |          | 0         | (base outcome) |       |       |           |

(sphus=Poor is the base outcome)

```

5 .      outreg2 using results_imp1_sloCountry_sup`IM'.xls, excel dec(3) alpha(0.001
> , 0.01, 0.05) symbol(**, **, *) ctitle(Slovenia2015)
results_imp1_sloCountry_sup.xls
dir : seeout

6 .
end of do-file

7 . do "C:\Users\hansg\AppData\Local\Temp\STD1bdc_000001.tmp"

8 .      log close The_value_of_a_helping_hand
name:    The_value_of_a_helping_hand
log:     C:\Users\hansg\Documents\working\SHARE data\weighting\output_1.smcl
log type: smcl
closed on: 10 Jun 2026, 09:25:27

```